# Design for fatigue strength

**Objective of fatigue calculations**

The reoccurrence of loads in machine components produces stresses, which are often called ***repeated, alternating*** **or *fluctuating* stresses**.
The objective of modern fatigue design is to ensure safe operation of a component during its entire life. This is achieved by using a combination of analytical methods, numerical simulations and data from experimental programmes.

- **Endurance or Fatigue limit**

This is the traditional approach. Assume that no fatigue damage is taking place below certain level of cyclic stress amplitude for a particular material.

- ***High cycle fatigue*** **(HCF)**

The number of load repetitions is large, typically over $10^3$ cycles and it is assumed that no initial defect exists in the structure. The cyclic stress level is generally below material yield. This approach is frequently called: **the stress-life or the nominal approach to fatigue**.

- ***Low cycle fatigue*** **(LCF)**

The number of load repetitions is small (less than $10^3$ cycles) and generally accompanied by a significant amount of plastic deformation. Cracks may initiate early in the life. The fatigue process is controlled by the strain deformation. This approach is frequently called: **the *strain-life* or the *local approach to fatigue*.**

- ***Fatigue crack growth*** **(FCG)**

A known defect assumed to exist in the structure and crack growth from the defect is related to the number of cycles using fracture analysis. It is usually suitable only for an elastic dominated stress and hence for HCF. This area is undergoing a rapid development during the last 15 years and is commonly referred to as.

The basics of high cycle fatigue (HCF) are given in the following sections while the area of FCG is covered during the introduction to fracture mechanics. Low cycle fatigue is beyond the scope of this course and is less used in the aerospace industry.

**Fatigue tests**

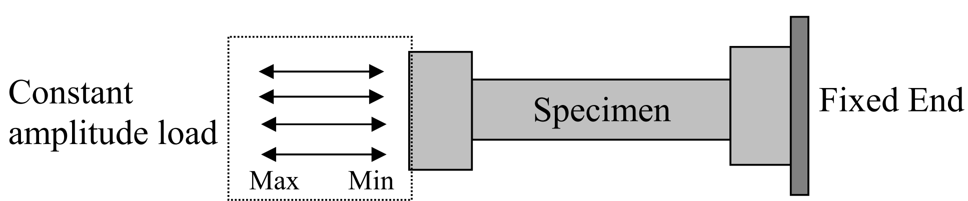
Fatigue analysis is based on data obtained in experiments. The basic data commonly used in fatigue strength design includes correlation between the **cyclic stress amplitude** and the **number of cycles to failure** of simple coupons.

However, likely deviations occur between the service components and the test piece due to many factors such as:

geometrical differences
loading conditions and
environmental effects, for example corrosion.

The simplest fatigue test takes a smooth piece of material and applies a fluctuating load, which varies between fixed upper and lower limits. Loading is continued until the specimen is separated (broken!) into two pieces. These tests are commonly called: **uniaxial, load control, constant amplitude tests**, and considered the base for the fatigue life estimation in the HCF region.

An example of such a test is shown below:



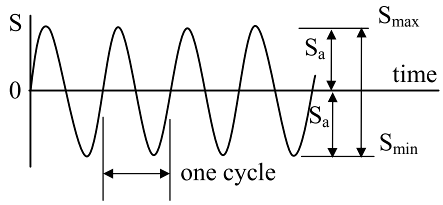

**Constant amplitude test**

This illustrates the simplest fatigue test. Most experimental programmes include test coupons that have notches, grooves or holes. For example, it is common practice in the aerospace industry to cycle to fracture a typical structural joint using two plates and a fastener.

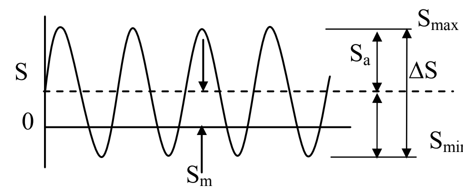
## Definitions

### <u>Fatigue under constant amplitude of applied stress, S (MPa)</u>

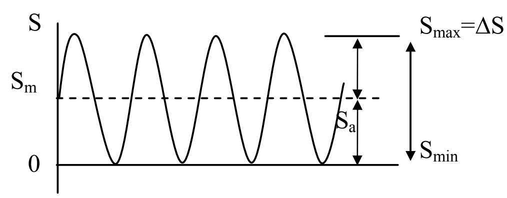
Three typical examples:




**a. Completely reversed, R= -1, $S_m$ = 0**




**b. Non-zero positive mean stress, $S_m$ >0**




**c. Tension only and $S_{min}$ = 0**

Assuming a constant cross section area the following cyclic stress definitions are used:

- The **<u>stress range, ΔS</u>** is the difference between the maximum and the minimum stress values.
- Averaging the maximum and the minimum stress values gives the **<u>mean stress, $S_m$</u>**.
- The mean stress $S_m$, may be zero, as in (a), but often it is not, as in (b).
- Half the range is called the **<u>stress amplitude, $S_a$</u>**, so that this is the variation about the mean.

$$\text{Stress range} \quad \Delta S = S_{max} - S_{min}$$

$$\text{Stress amplitude } S_a = \frac{\Delta\sigma}{2} = \frac{S_{max} - S_{min}}{2}$$

$$\text{mean stress } S_m = \frac{S_{max} + S_{min}}{2}$$

The term **alternating stress** has the same meaning as **stress amplitude**. It is also useful to note that:

$$\boxed{S_{max} = S_m + S_a \quad \text{and} \quad S_{min} = S_m - S_a}$$

The signs of $S_a$ and $\Delta S$ are always positive, since $S_{max} > S_{min}$, where tension is considered positive. The quantities $S_{max}$, $S_{min}$, and $S_m$ can be either positive or negative. The ratio between minimum and maximum stress, R, is often used:

$$R = \frac{S_{min}}{S_{max}} \quad \text{and therefore:}$$

$$S_m = \frac{S_{max}}{2}(1+R) \quad \text{and} \quad \Delta S = S_{max}(1-R)$$

where **R** is called the **stress ratio** or **R-ratio**.

- Cyclic stress with **zero** mean can be specified by giving the amplitude $S_a$, or by giving the numerically equal maximum stress, $S_{max}$.

- The term ***completely reversed stress*** is used to describe a situation of $S_m = 0$, hence $R = -1$, as in figure (a) on previous page. Zero-to-tension stressing refers to cases of $S_{min} = 0$, hence $R = 0$, as in figure (c) on previous page.

- If the **mean stress is not zero**, two independent values are needed to specify the loading. Some combinations that may be used are: $S_a$ and $S_m$, $S_{max}$ and R and $\Delta S$ and R, $S_{max}$ and $S_{min}$.

The same system of subscripts and the prefix $\Delta$ are used in an analogous manner with other variables, such as reversed bending moments M, cyclic strains $\varepsilon$, fluctuating load (force) P.
For example, $P_{max}$ and $P_{min}$ are maximum and minimum loads, $\Delta P$ is load range, $P_m$ is mean load, and $P_a$ is load amplitude. If there is any possibility of confusion as to which variable is used with the ratios R, a subscript should be used, such as $R_\varepsilon$ for strain ratio.

Note: ***FOR LOCAL STRESS CALCULATION S IS COMMONLY REPLACED BY $\sigma$***

## Historical Fatigue Example – the Comet jet aeroplane failure

The Comet jet history demonstrates the importance of design for fatigue in the commercial jet transportation.

**Background**

- It was the first modern commercial jet design after WW2.
- The first commercial plane designed to cruise at high altitude of 12000m.
- Therefore the first plane where the fuselage was pressurised and depressurised during every flight to maintain passengers' comfort.
- Fuselage had to be capable of withstanding the fatigue stress cycles.
- It is widely believed that the Comet fatigue failure set back the British industry and assisted in the emergence of Boeing as a commercial leader.

**Sequence of disasters**

- 2 May, 1953 – Comet plane disintegrates in mid-air soon after take off in India during a heavy tropical thunderstorm. Investigation concludes the accident is due to extreme stormy weather and crew control. It is not viewed as a cause for concern.
- 10 January, 1954 – A comet explodes over the Mediterranean Sea in good weather. No flaws in design discovered and the plane remain in service.
- 8 April, 1954 – third crash above Rome investigation begins and the plane is grounded but no real reason has been found.

**Investigation**

- A testing set up is constructed using a Comet plane on the ground: the plane fuselage is subjected to repeated pressurised cycles and the wings to the cyclic flight loads.
- After about 3000 pressurisation cycles fatigue cracks initiate and grow from a corner of a window and advanced until…..
- Secondary cracking take place by bending and shearing off the center over the outer portion.
- Complete piercing and catastrophic failure.

## Conclusions

The failures of the three Comet jet planes were due to fatigue cracking that was not accounted for in the initial plane design. The design was modified and in the subsequent planes the windows sections were replaced with a new reinforced panel.

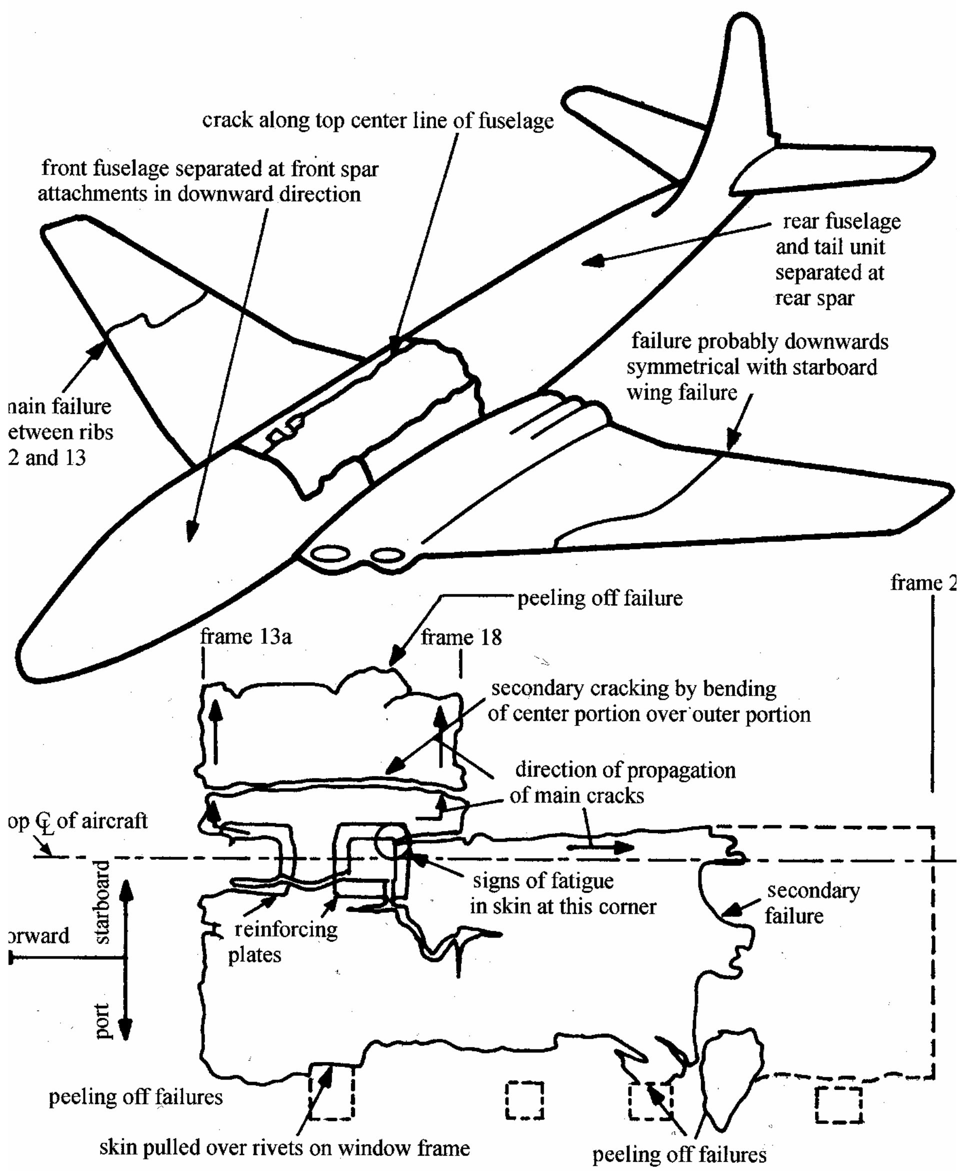



Fig. 1.1. Schematic diagram illustrating the location of fatigue cracks in a failed Comet airplane. (After Petroski, 1996.)

**Some basic considerations in fatigue calculations:**

| Aspect | Testing of Laboratory specimens | In - Service components |
|---|---|---|
| Loading | Uniaixial, constant amplitude | Irregular, multiaxial, not-completely known |
| Geometry | Small, smooth specimen | Complex with notches, grooves fillets etc. |
| Environment | Well known and controlled | Corrosion, temperature, fretting |
| Manufacturing | Good surface finish, homogeneous, heat treated | As received, surface treatment etc. |
| Scatter | Scatter usually small | Scatter usually large |
| Material microstructure | Inspection is carried out during the tests | Only snap-shots may be possible |

**Fatigue lecture objectives:**

1. Understand basics fatigue failure and the fatigue terminology.
2. Prediction of fatigue strength and fatigue life using the S-N diagram for both, finite and infinite life.
3. Be able to account for stress concentration, mean stress and simple multiaxial load.
4. Apply the stress-life analysis to design.
5. Calculate service component fatigue damage using damage summation.

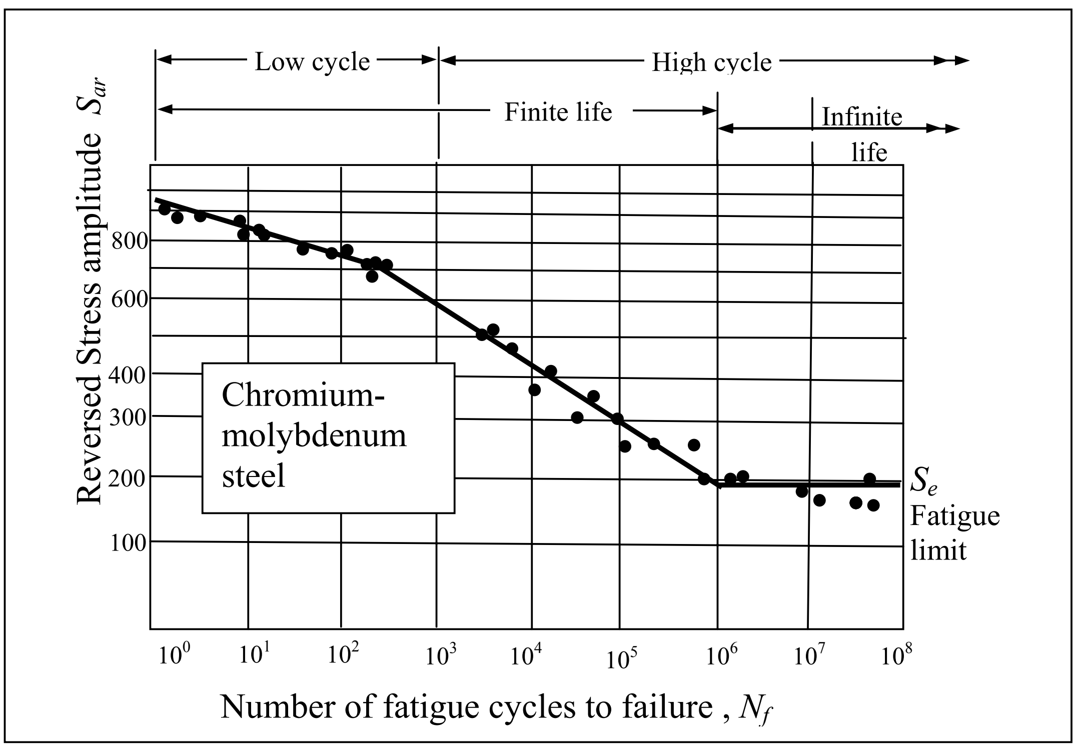
# Fatigue using Stress-life approach (High Cycle Fatigue)

The stress-life approach is the traditional and still the most widely used method of estimating the life of components. It has a long history, initially developed during the industrial revolution of the 19th century when metal structures became widely used in engineering design.

## 1. Fatigue limit and the S-N diagram

The first concept that emerged from a stress-life analysis of several metals was that of '**endurance limit**', which characterises an applied stress amplitude below which the material is expected to have an **infinite life**. Using this concept for design, it is therefore only necessary to ensure that the stress in the structure is below the **material fatigue limit**. However, this simplistic approach may be inappropriate in many fatigue cases.

To establish the strength of materials under fatigue loads, a series of tests are conducted on similar specimens subjected to fluctuating**, completely reversed stress, R=-1 and mean stress is zero!**. The number of cycles to failure of the specimens is recorded and the stress amplitude is plotted on a log paper against the number of cycles to failure. A typical S-N diagram of steel alloy is shown below.



The S-N diagram has two distinct regions:

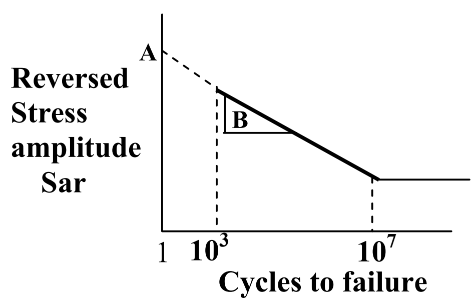
**Finite life** $\longrightarrow$ **Number of cycles $10^7$ or less** $\longrightarrow$ **Specimens failed**

**Infinite life** $\longrightarrow$ **Number of cycles $10^7$ or over** $\longrightarrow$ **Specimen do not fail**

It is customary in design applications and standards to assume that no fatigue failure may occur in a component if the **stress amplitude**, called the **fatigue strength**, corresponds to a number of cycles above $10^7$ cycles. This stress is often referred to as the material '**endurance limit**' or '**fatigue limit**' explained above.

However, some structural steels such as non-ferrous metals and alloys **do not have an endurance limit!**

Within the S-N diagram, the data in the region of life to failure between $10^3$ and $10^7$ cycles is found to approximate a straight line on a log-log plot. The corresponding relation is illustrated below.




$$S_{ar} = A\, N_f^{\,B}$$

$$10^3 < N_f < 10^7$$

**$S_{ar}$ is the <u>reversed stress amplitude</u> or the <u>cyclic stress</u> at <u>R = -1</u> (zero mean stress) and is the stress value shown on the material S-N diagram.**

The finite life region of the S-N diagram could also be split into the **high cycle fatigue life** and the **low cycle fatigue life**. In the low cycle fatigue we assume **material yield** at the critical area and therefore a stress based approach may not be appropriate.

The exact borderline between the LCF and the HCF regions is not always very clear but is most often assumed to be at fatigue strength corresponding to a life of between $10^3$ to $10^4$ cycles.

Past design books provided factors to modify the endurance limit for practical application by, for example, introducing factors that account for each single effect. These factors were used to modify (usually reduce) the endurance limit of the laboratory controlled test components. For instance:

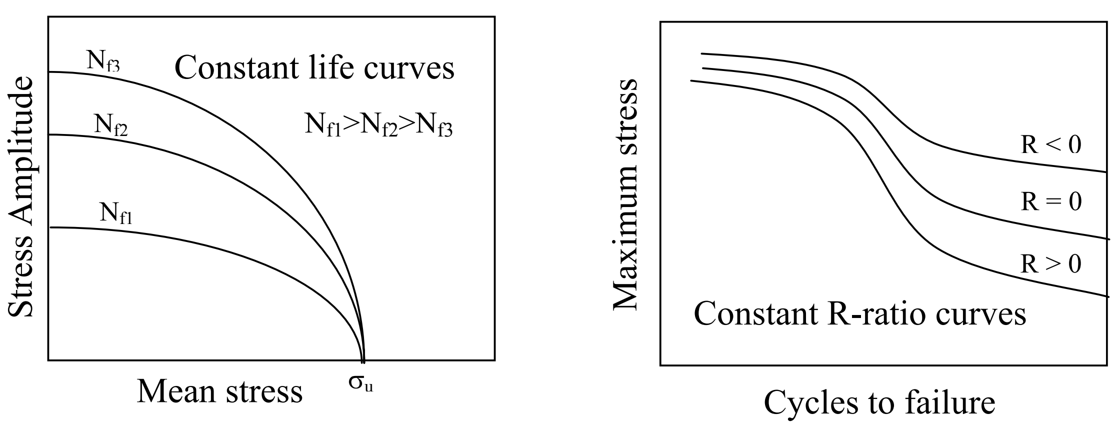
$$S_e = K_a K_b K_c K_d K_e K_f \ldots\ldots S'_e$$

Where $S_e$ is the modified endurance limit. $K_{a,b,c,d,e,f\ldots}$ are the modifying parameters for different effects, of for example; manufacturing, geometry, reliability, temperature etc., taken from tables and graphs. However, this simplified approach could lead to errors and is not widely used anymore.

## 2. Mean stress effect

So far we have only mentioned fully reversed cyclic stress where the mean stress of the fatigue cycle is zero, $S_m$=0. In many applications this is not representative and the mean stress has shown to influence the fatigue behaviour of metals.

There are several ways to include the mean stress effect in fatigue life procedures. One popular method is the ***constant life diagram***. This involves plotting values of several combinations of mean stress and stress amplitude that produce the same life to failure, each curve indicating a constant life. Another procedure often used is to plot curves of the ***Constant R-ratio*** ($R = S_{min}/ S_{max}$ ) for the maximum stress versus life to failure.



**Mean stress correction models**

Several models were developed to compare life plots of tests using different mean stress. These models are primarily used to estimate the life of a component with a particular mean stress to the fully reversed (zero mean stress) material cyclic test data.
A frequently used relationship is:

$$\frac{S_a}{S_{ar}} + \left(\frac{S_m}{S_u}\right)^n = 1$$

Where:

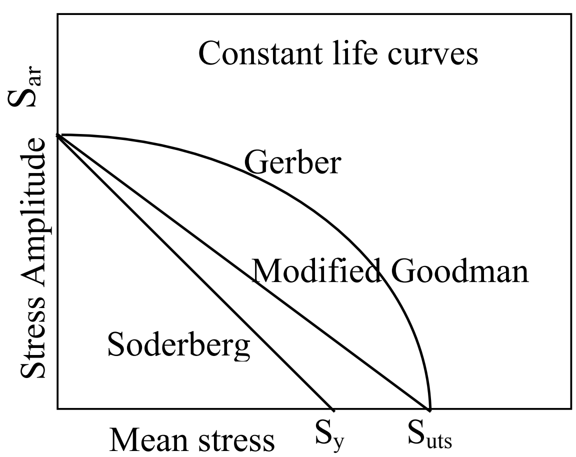
Sa = Applied stress amplitude (with mean stress)
Sm = Mean stress
Su = Ultimate tensile stress
Sar = Reversed stress amplitude (no mean stress)

In the equation above, 'n' value refers historically to the following:

1. n = 1 The **'modified Goodman equation'**
2. n = 2 The **'Gerber parabola'**
3. n = 1, $S_u = S_y$ The **'Soderberg relation'**

$S_u$ is the material ultimate strength and $S_y$ is the yield stress.

The above three models could be presented graphically on the constant life diagram by plotting the mean stress Vs the stress amplitude, where $\sigma_{ar}$ is the stress amplitude in the case of zero mean stress (R = -1):



Other relations also exist between the fully reversed data and the mean stress such as the '**Walker equation**' and corrections that account for **negative** mean stress.

The following observations are usually made about the foregoing mean stress models:

1. Soderberg provides a conservative estimate of fatigue life.
2. Goodman is adequate for brittle materials but conservative for ductile behaviour if tension dominate the fatigue life.
3. Gerber is in general good for ductile materials such as steel alloys but it does not distinguish between life in tensile or compression mean stress.

**If not mentioned otherwise, the modified Goodman is considered for mean stress correction in these lectures.**

**Example (mean stress)**

Calculate the fatigue life of a smooth bar subject to a constant amplitude cyclic stress range of 1110.6 MPa and mean stress of 200 MPa. Materials properties are: $S_u$ = 1400, $S_y$ = 1170, A = 1240 all in MPa and B = -0.06.

Solution:

$$S_a = \frac{\Delta S}{2} = \frac{1110.6}{2} = 555.3 \text{ MPa}$$

Using modified Goodman:

where: $\frac{S_a}{S_{ar}} + \left(\frac{S_m}{S_u}\right)^1 = 1$

where: $S_{ar} = AN_f^B$

$$S_{ar} = \frac{S_a}{1 - \frac{S_m}{S_u}} = \frac{555.3}{1 - \frac{200}{1400}} = 647.85 \text{ MPa}$$

$$N_f = \left[\frac{S_{ar}}{A}\right]^{\left(\frac{1}{B}\right)} = \left[\frac{647.85}{1240}\right]^{\left(-\frac{1}{0.06}\right)} = 50\text{x}10^3 \text{ Cycles to failure}$$

*(Hand calculators hint: invert the fraction to use a positive exponent)*

Repeat this example using Soderberg and compare the results

## 3. Stress concentration effect

One of the major factors of reduction in life of a service part in comparison to tests data is the change in geometry that locally raises the cyclic stress. These unavoidable geometric discontinuities also affect the local stress field and the shape and form of the fatigue cracks. Much work was done in the past to compare the so called notched parts to smooth data. We have here restricted our study to the classical approach to fatigue.

In **static design** the effect of notches is considered by using engineering handbooks that contain data of different notch geometry and the relevant stress concentration factor (SCF) and applying:

$$\boxed{\sigma = K_t S}$$

Where **$\sigma$ is the local stress** in the vicinity of the notch; **S the global or NOMINAL stress** and **$K_t$ is the stress concentration factors** taken from data books.

In fatigue the SCF is replaced by the so-called **Fatigue notch factor, $K_f$** (sometimes also called **fatigue strength reduction factor**):

$$\boxed{K_f = \frac{\sigma}{S} = \frac{\text{Cyclic stress with the notch}}{\text{Cyclic stress without notch}}}$$

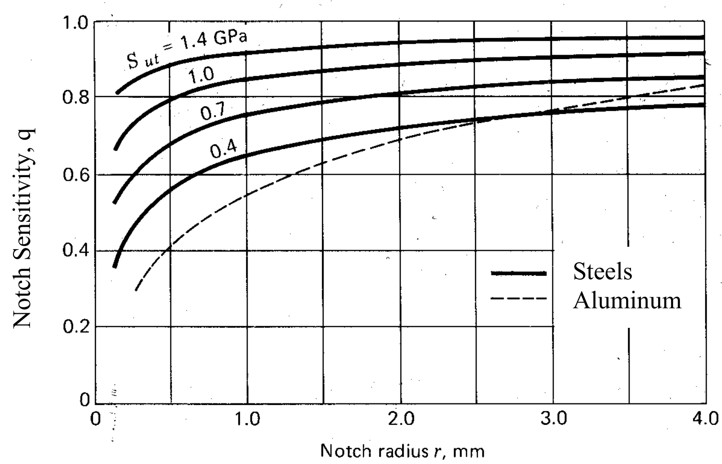
It was shown that in most fatigue cases that reduction in fatigue strength is less than the increase in the local static stress, which means that $K_f < K_t$ in most cases.

A concept was introduced to the fatigue of notched components using a **notch sensitivity factor, q**. This factor is used to correlate between $K_t$ and $K_f$ and is mainly a **material parameter**.
Various relationships were developed between these three parameters ($K_t$, $K_f$ and q). Most of them use empirical (experimental) relation. The most widely used ones take the form:

$$\boxed{q = \frac{K_f - 1}{K_t - 1}}$$

Since $K_t$ is widely available for many notch geometries and q is a given material data, it is possible to calculate $K_f$, the fatigue reduction factor.

An example of a notch sensitivity graph for various notch radii is given below (Shigley, Mechanical Engineering Design, 1977) for four steels and an aluminium alloy:



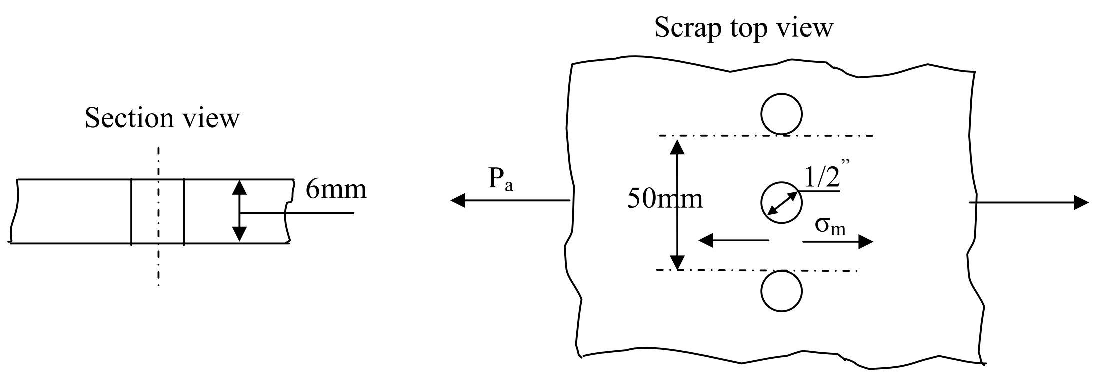

In engineering literature, the value of q is sometimes represented using the following empirical formula:

$$\boxed{q = \frac{1}{1 + \frac{\alpha}{r}}}$$

Where r is the notch radius (in mm) and α is a material constant.

**Example: Fatigue life estimates - stress concentration and mean stress**

Calculation of fatigue life is required for the top skin joint on the wing of a large commercial jet. The joint skin plate is made of standard aluminium Al 2024-T3. From a separate load analysis it was found that a remote cyclic reversed load of about 20 kN and a local constant stress of approximately 70 MPa is applied to the structure. Estimate the fatigue life of the plate in the vicinity of the bolt's hole as shown below. The 2024-T3 properties are: S-N curve constants, A=839 MPa, B=-0.102, $\sigma_u$=476 MPa and $\alpha$=0.51. Assume factor of safety of 1.8.




<u>Solution procedure</u>:

1. It is first necessary to obtain the notch concentration factor, $K_t$, and the notch sensitivity, q using the available data.
2. From those, the fatigue notch factor, $K_f$ is calculated.
3. Next, the cyclic and mean stress components' are evaluated.
4. One of the mean stress correction methods is used to obtain the equivalent reversed stress amplitude, $\sigma_{ar}$ and this is multiplied by $K_f$ to obtain the local fatigue stress amplitude.
5. This reversed stress is used to estimate the life from the material S-N curve.

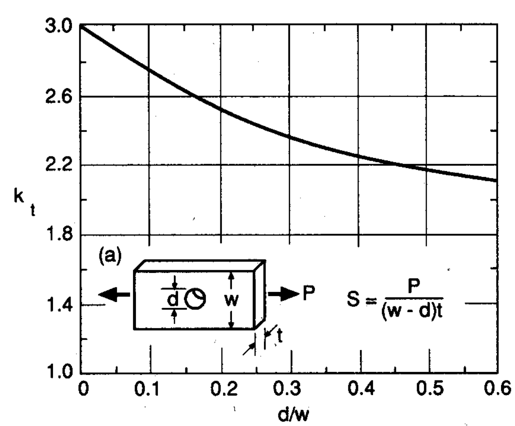
**Solution:**

**1**. Using the figure shown below for stress concentration factors (Peterson 1974):

W = 50mm
d = 12.7mm
r = 6.35mm
d/W = 0.254

↓

**Kt = 2.4**



The notch sensitivity is: $$q = \frac{1}{1 + \frac{0.51}{6.35}} = 0.926$$

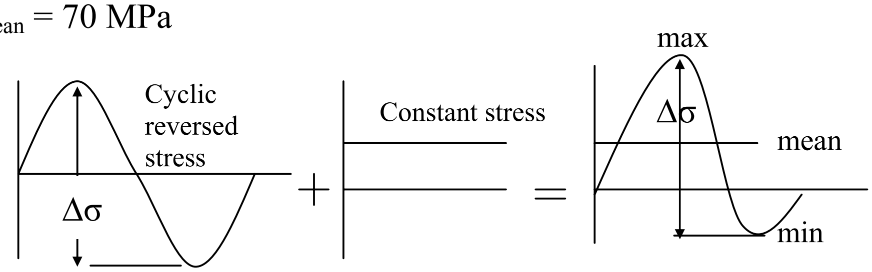
**2**. And hence the fatigue notch factor:

$$K_f = 1 + q(K_t - 1) = 1 + 0.926(2.4 - 1) = 2.296$$

**3**. We now obtain the applied cyclic stresses:

$$\Delta\sigma_{\text{local cyclic range}} = \left[\frac{P}{(w-d)t}\right] = \left[\frac{20x\,10^3}{(50-12.7)6}\right] = 89.4 \text{ MPa}$$

$$\sigma_{constant} = \sigma_{mean} = 70 \text{ MPa}$$




$$\sigma_a = \frac{\Delta\sigma}{2} = \frac{89.4}{2} = 44.7 \text{ MPa}$$

$$\sigma_{mean} = \sigma_{cons\tan t} = 70 \text{ MPa}$$

**4**. The completely reversed stress using the modified Goodman theory (FOS=1.8):

$$\sigma_{ar} = \frac{\sigma_a}{1 - \frac{\sigma_m}{\sigma_u}} = \frac{44.7}{1 - \frac{70}{476}} = 52.4$$

And: $\underset{(Sar)}{52.4}$ x $\underset{(FOS)}{1.8}$ x $\underset{(Kf)}{2.296}$ = 216.6 MPa

**5**. Finally using the material S-N curve equation:

$$N_f = \left[\frac{\sigma_{ar}}{A}\right]^{\left(\frac{1}{B}\right)} = \left[\frac{216.6}{839}\right]^{\left(-\frac{1}{0.102}\right)}$$

$$N_f = 583{,}407 \text{ cycles}$$

**Example (in class exercise – stress concentration)**

A shaft with a step-down in diameter has dimensions, as defined in Figure 1 below, of $d_1 = 50$, $d_2 = 55$, and $\rho = 1.3$mm. The shaft is subjected to reversed bending and is made of a quenched and tempered steel. The $10^6$ cycles fatigue strength in bending of this steel using smooth specimens is $\sigma_{ar} = 500$ MPa.
What completely reversed bending moment $M_a$ can be applied to the notched shaft for $10^6$ cycles before fatigue failure expected? (use Figures 1 and 2).

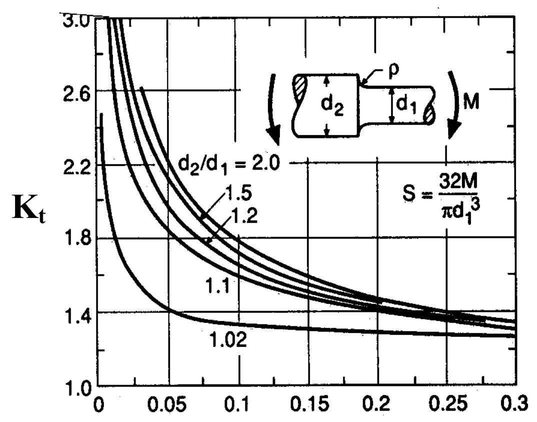
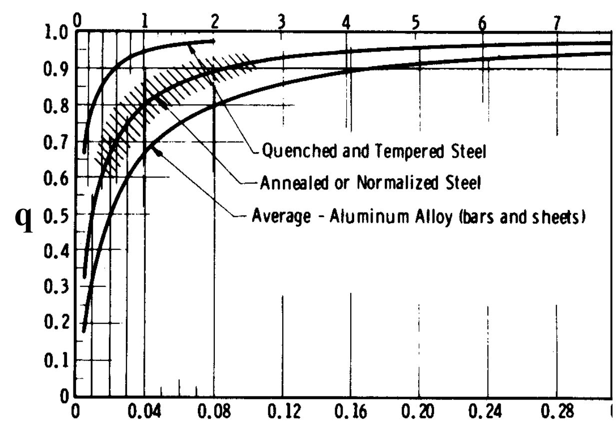



Figure 1

**Notch radius, ρ, in mm**




**Notch radius, ρ, in inches**

Figure 2

Solution procedure: obtain $K_t$ and q, calculate $K_f$, calculate allowed fatigue strength, and calculate $M_a$ using bending theory.

**Example (in class exercise – mean stress)**

A shaft of circular cross section is subjected to a steady bending moment of 1500 Nm and simultaneously to an alternating bending moment of 1000 Nm in the same place. Calculate the necessary diameter of the shaft to withstand fatigue failure if a factor of safety of 2.5 is to be used. The material yield stress is 210 MPa (use Soderberg rule) and the fatigue limit in reversed bending is 170 MPa.

Solution procedure:

a. Determine the mean and amplitude moments.
b. Determine the mean and amplitude stresses using bending theory, diameter unknown.
c. Set up the Soderberg mean stress correction equation, include the FOS. Resolve for the diameter.

## 4. Irregular load history

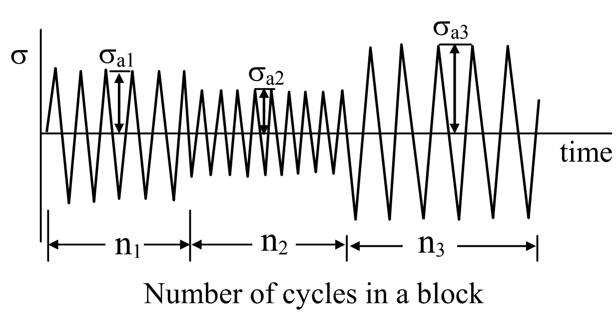
In practice, engineering components are rarely subjected to constant load amplitude and airframes are no exception. Methods and standards have been developed that compare between the irregular load cycles fatigue damage and the constant amplitude laboratory coupon results. This allows an estimate of the service component damage providing that some form of the load history is available.

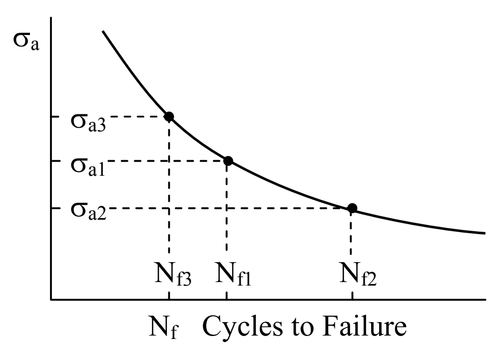
### Damage summation

A simple and widely used criterion for predicting the extent of fatigue damage induced by a block of cycles having the same stress amplitude and mean is called the **Miner's rule.** It implies that the fraction of life used by a certain block of cycles is equal to the ratio between the number of cycles used in the block, $n_i$, and the number of cycles to failure, $N_f$, for the same stress amplitude, on the S-N curve. Failure is expected when the total sum of all such fractions is equal to unity.

$$\boxed{\frac{n_1}{N_{f1}} + \frac{n_2}{N_{f2}} + \frac{n_3}{N_{f3}} + \ldots\ldots = \sum_{i=1}^{i=m} \frac{n_i}{N_{fi}} = 1}$$

This is illustrated below:



It is often customary for practical applications to estimate the damage for a particular service history, for example an aeroplane's particular flight pattern that includes **taxiing – take off – cruising - landing**. The sequence is repeated until the cumulative damage reaches unity and the **number of repetitions** to failure is calculated:

$$\boxed{\text{Number of repetitions} = \frac{1}{\sum_{i=1}^{i=m} \frac{n_i}{N_{fi}}}}$$

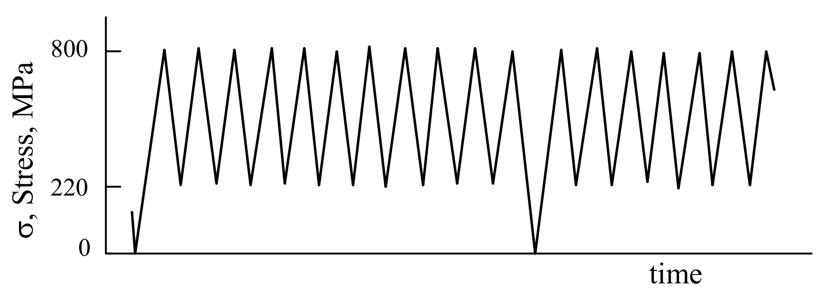
Although the Miner's rule is widely used it is essential to understand its limitations, in particular:

1. It is a LINEAR summation model where cycles at different stress amplitude have the same weight on the damage and total failure.
2. The ORDER in which the stress blocks are applied has no significance.
3. Fatigue failure occurs when the sum of the fractions reaches a particular value – usually one.

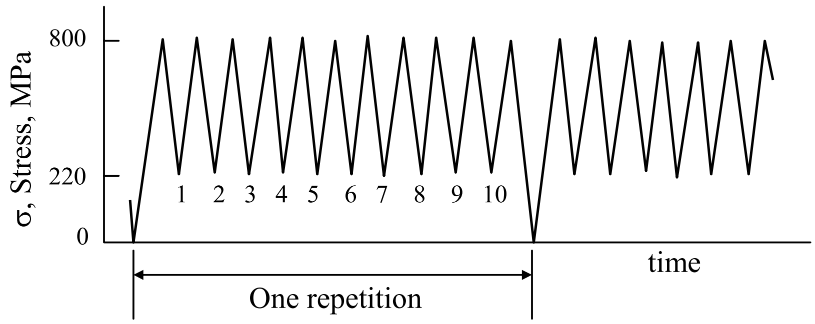
It has been shown and is widely accepted that these three assumptions are often inaccurate.

**Example - irregular load 1**

The stress history shown below was measured near a titanium Ti-6Al-4V landing gear component during normal operational conditions. Estimate the number of repetitions to failure of this stress history if it has the following fatigue properties: $\sigma_u$ = 1233 MPa, A = 1889 MPa and B = -0.104.



Solution:




In one repetition of the history we count one large cycle and 10 smaller ones.

For the large cycle the stress amplitude and mean are:

$\sigma_a = (800 - 0)/2 = 400$ MPa and $\sigma_m = (800+0)/2 = 400$ MPa

Using the modified Goodman correction to obtain the fully reversed stress amplitude:

$$\sigma_{ar} = \frac{\sigma_{a,effective}}{1 - \dfrac{\sigma_{m,effective}}{\sigma_u}} = \frac{400}{1 - \dfrac{400}{1233}} = 592.1 \text{ MPa}$$

and from the material S-N curve the number of cycles to failure is:

$$N_f = \left[\frac{\sigma_{ar}}{A}\right]^{\left(\frac{1}{B}\right)} = \left[\frac{592.1}{1889}\right]^{\left(-\frac{1}{0.104}\right)}$$

$$N_f = 69917 \text{ cycles}$$

For the smaller stress cycles similar analysis:

$\sigma_a = (800 - 220)/2 = 290$ MPa and $\sigma_m = (800+220)/2 = 510$ MPa

$$\sigma_{ar} = \frac{\sigma_{a,effective}}{1 - \dfrac{\sigma_{m,effective}}{\sigma_u}} = \frac{290}{1 - \dfrac{510}{1233}} = 494.6 \text{ MPa}$$

$$N_f = \left[\frac{\sigma_{ar}}{A}\right]^{\left(\frac{1}{B}\right)} = \left[\frac{494.6}{1889}\right]^{\left(-\frac{1}{0.104}\right)}$$

$$N_f = 394677 \text{ cycles}$$

Summation for one repetition using Miner's rule:

$$\text{Number of repetitions} = \frac{1}{\sum_{i=1}^{i=m} \dfrac{n_i}{N_{fi}}} = \frac{1}{\left(\dfrac{1}{69917}\right) + \left(\dfrac{10}{394677}\right)} = 25227 \quad \text{(Until failure)}$$

25227 repetition of this load history to failure assuming Miner's rule damage summation.

(Or fraction of fatigue damage for each repetition is $3.964 \times 10^{-5}$)

**Example - irregular load 2**

Two smooth steel bar specimens were subjected to fatigue tests under cyclic axial stress of ±400 MPa and ±250 MPa. Specimens failure occurred after $2x10^4$ and $1.2x10^6$ cycles respectively, at these two tress levels. Estimate the fatigue life of a similar specimen subjected to a stress level of ±300 MPa if it has already undergone $2.5x10^4$ cycles at ±350MPa.

Solution:

First we calculate the S-N curve constants A and B from the two tests

We have:
$S_{a1} = 400$ MPa    $N_{f1} = 2x10^4$
$S_{a2} = 250$ MPa    $N_{f2} = 1.2x10^6$

New test:
$S_{a3} = 350$ MPa    $n_3 = 2.5x10^4$ (already used)
$S_{a4} = 300$ MPa    $n_4 = ?$ (left until failure)

$$S_a = A\, N_f^{\ B}$$

Since $A_1 = A_2$

$$\frac{S_{a1}}{S_{a2}} = \left(\frac{N_{f1}}{N_{f2}}\right)^B \qquad \frac{400}{250} = \left(\frac{2x10^4}{1.2x10^6}\right)^B \qquad 1.6 = \left(\frac{1}{60}\right)^B$$

$$B = \frac{\log 1.6}{\log\left(\frac{1}{60}\right)} = -0.1148 \qquad A = \frac{S_{a1}}{N_{f1}^B} = \frac{400}{\left(2x10^4\right)^{-0.1148}} = 1247\ \text{MPa}$$

The expected failure from the first part of the test at stress amplitude of 350 MPa is

$$N_{f3} = \left[\frac{S_{a3}}{A}\right]^{\left(\frac{1}{B}\right)} = \left[\frac{350}{1247}\right]^{\left(-\frac{1}{0.1148}\right)} = 6.4 \mathrm{x} 10^4 \text{ cycles}$$

and the part damage is $\frac{n_3}{N_{f3}} = \frac{2.5 \mathrm{x} 10^4}{6.4 \mathrm{x} 10^4} = 0.39$

The expected failure from the second part of the test at stress of 350 MPa is

$$N_{f4} = \left[\frac{S_{a4}}{A}\right]^{\left(\frac{1}{B}\right)} = \left[\frac{300}{1247}\right]^{\left(-\frac{1}{0.1148}\right)} = 24.5 \mathrm{x} 10^4 \text{ cycles}$$

And finally applying Miner's rule

$$\frac{n_3}{N_{f3}} + \frac{n_4}{N_{f4}} = 1 \qquad N_4 = (1 - 0.39) 24.5 \mathrm{x} 10^4 = 1.49 \mathrm{x} 10^5 \text{ cycles}$$

until failure is expected in the 4th incomplete test.

# Tutorials STRESS - LIFE

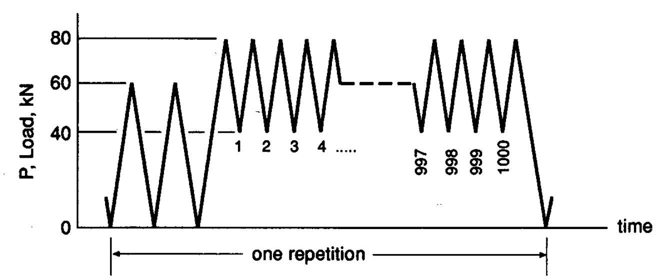
1. The analysis of the cyclic stresses on steel part of an undercarriage aircraft component shows that during each flight it is subjected to the following stress history: $N_1$ = 100 cycles at +/- 400 MPa and $N_2$ = 10 cycles at +/- 500 MPa. If A and B are 1600 MPa and –0.1 respectively, estimate the component life before failure assuming total flights = $1/(N_1/N_{1f}+ N_2/N_{2f})$. (5429 flights).

2. A stepped shaft with a smallest diameter D = 15mm is subjected to the load history shown below. Assume that the S-N curve with a fatigue notch effect of $K_f$= 1.92 included is:

$$S_{(R=-1)} = 2250\, N_f^{-0.172} \text{ MPa}$$

A factor of safety of 10 on the life (repetitions) is required and 50 load history repetitions are anticipated and the material has ultimate stress $\sigma_u$ = 1000 MPa.

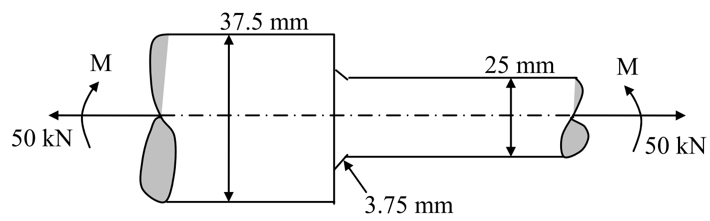
Estimate the number of repetitions to failure of the load history. (307).




3. A steel bar is subjected to a fluctuating axial load that varies from a maximum of 330 kN in tension to a minimum of 100 kN in compression. The steel properties are: $\sigma_u$ = 1090 MPa, $\sigma_y$ = 1010 MPa and $\sigma_e$ = 510 MPa (fatigue limit). Determine the bar diameter to give infinite life based on factor of safety of 2.5. (Answer: 38.7mm).

4. The stepped shaft shown in the figure below is subjected to a steady axial pull of 50 kN and a uniform bending moment M. If the yield strength of the shaft material is 300 MPa and the fatigue limit in reversed bending is 200 MPa,

calculate the maximum value of M to avoid fatigue failure in the shaft. $K_t$ for the fillet radius is 1.55 and the notch sensitivity factor q=0.9. (135.1 Nm)

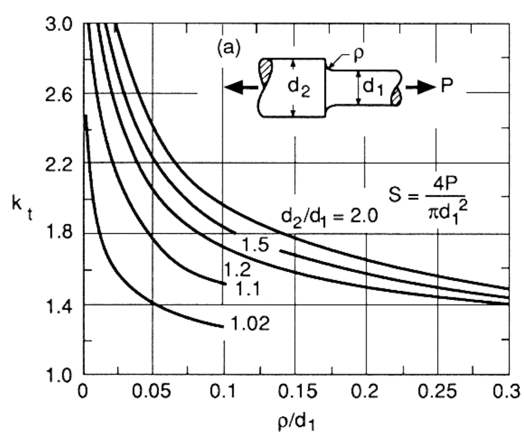



5. For an S-N curve of the form $\sigma_a = AN_f^B$, two points ($N_1$, $\sigma_1$) and ($N_2$, $\sigma_2$) are known. Develop equations for the constants A and B as a function of these values. Considering the S-N plot on a log-log coordinates, what are the significance of A and B?

6. A circular rod made of aircraft quality steel ($\sigma_u$ = 1172 MPa, A=1643, B= -0.0977 and q = 0.9) is loaded axially and has a step change in diameter. The dimensions (see the figure overleaf (Peterson, 1980) are, $d_1$ = 15, $d_2$ = 18 and $\rho$ = 1 mm.

   a. Calculate safety factors in both, stress and life if during service 30000 cycles are applied at zero-to-tension load, where the peak load is 70 kN (1.165, 4.74).

   b. If a stress safety factor of 1.7 is required. Is the design adequate? If not, try to find a suitable fillet radius.



# Fatigue Advanced Topics

## Multiaxial stresses effect

So far, our analysis has been restricted to the uniaxial cyclic stress that is in the same direction as the applied cyclic load. However, most engineering components are subjected to complex loading and the local stress is not uniaxial, but multiaxial. It involves more than one stress component. Many theories were developed and are still under research to try to predict parts' lives due to multiaxial loading.

Most of the multiaxial fatigue approaches introduce effective cyclic and mean stresses that could be compared with the uniaxial S-N curve. One such approach is to assume that the effective stress is related to the *octahedral shear stress yield criteria*:

$$\boxed{\sigma_{a,efective} = \frac{1}{\sqrt{2}}\sqrt{\left(\sigma_{xa}-\sigma_{ya}\right)^2 + \left(\sigma_{ya}-\sigma_{za}\right)^2 + \left(\sigma_{za}-\sigma_{xa}\right)^2 + \left(\tau^2{}_{xya} + \tau^2{}_{yza} + \tau^2{}_{zxa}\right)}}$$

Where $\sigma_{a,effective}$ is the *effective stress amplitude* and $\sigma_{xa,ya,za}$ are the cyclic stress amplitude components.

Using this criteria the effective mean stress, $\sigma_{m,effective}$ is the sum of the three mean stress components:

$$\boxed{\sigma_{m,efective} = \sigma_{xm} + \sigma_{ym} + \sigma_{zm}}$$

To obtain a completely reversed (R=-1) cyclic stress, a mean stress theory can be used, for example the modified Goodman correction:

$$\boxed{\sigma_{ar} = \frac{\sigma_{a,effective}}{1 - \dfrac{\sigma_{m,effective}}{\sigma_u}}}$$

In the special case of torsion fatigue the octahedral shear stress is:

$$\boxed{\sigma_{a,efective} = \sqrt{3}\,\tau_{xya} \qquad \sigma_{m,efective} = 0}$$

Example (Multiaxial fatigue including mean stress)

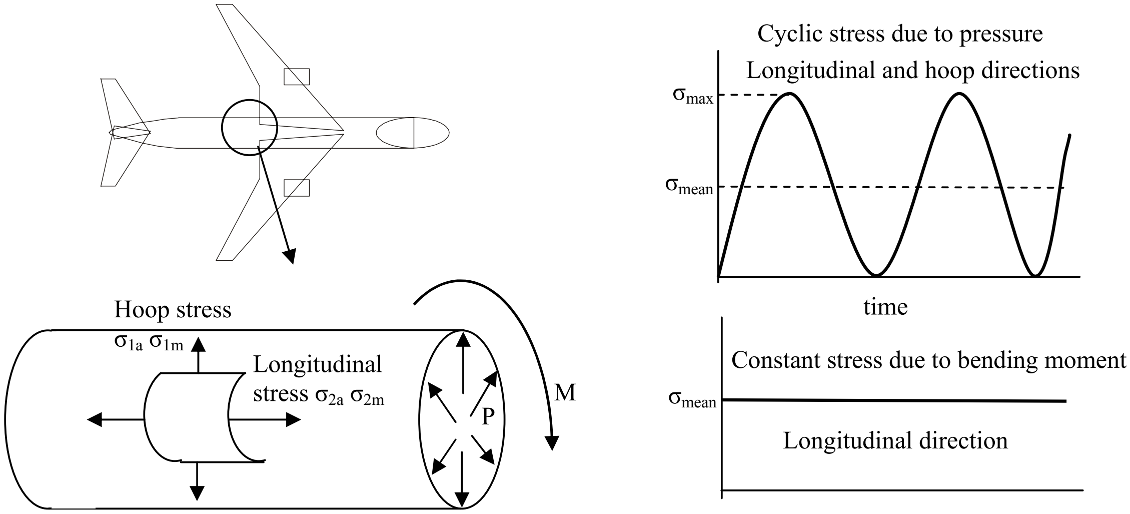
The fuselage of a medium commercial jet is pressurised and depressurised upon every takeoff and landing. It has a life expectancy of 25 years with an average of 3 takeoffs a day. The fuselage is made of aluminium 2024-T3 and has an approximate diameter of 4m and a wall thickness of 1”. The maximum internal pressure during flight is about 3.9 MPa. Due to the combined fuselage and tail weights, part of the fuselage is also subjected to a maximum constant bending moment of 3 MNm.

What is the expected life in flight cycles of the plane fuselage and how is this compared with an estimated life, using the material S-N curve and assuming a minimum fatigue safety factor of 2? (The 2024-T3 properties are: S-N curve constants, A=839 MPa, B=-0.102, $\sigma_u$=476 MPa and $\alpha$=0.51).

**Solution:**

We consider the fuselage to be a thin walled cylinder under cyclic internal pressure and constant bending moment. In this case the principal stresses coincide with the hoop and longitudinal stress directions and hence the shear stress components are zero. We also consider the radial, through the wall thickness stress to be zero (plane stress).

The stress due to the internal pressure fluctuates between 0 - maximum while the stress due to the bending is a constant tension. The pressure contributes to longitudinal and hoop stress components, mean and alternating, and an additional longitudinal mean component exists due to the bending moment.




The problem is solved by using:

- Multiaxial stresses – an effective octahedral shear stress.
- Modified Goodman for mean stress.
- Material S-N curve.
- Note that fatigue and not strength safety factor is used.

The number of cycles during the plane operation is:

365 x 3 x 25 = 27,375 flight cycles

The stress components are:

$$\sigma_{1a} = \sigma_{1m} = \left( \frac{\frac{P_{max}}{2} r}{t} \right) = \frac{3.9 x 2 x 10^3}{2 x 25.4} = 153.54 \text{ MPa}$$

(Hoop due to pressure)

$$\sigma_{2a} = \left( \frac{\frac{P_{max}}{2} r}{2t} \right) = 76.77 \text{ MPa}$$

(Longitudinal due to pressure)

$$\sigma_{2m} = \left( \frac{\frac{P_{max}}{2} r}{2t} \right) + \frac{M_{max} r}{I} = 76.77 + \frac{M_{max}}{\pi r^2 t} = 76.77 + \frac{3000 x 10^6}{\pi \left(2x10^3\right)^2 25.4} =$$

$$76.77 + 9.4 = 86.17 \text{ MPa}$$

(Longitudinal due to pressure and moment)

$$\sigma_{3a} = \sigma_{3m} = 0 \qquad \text{(plane stress)}$$

The effective octahedral shear amplitude is:

$$\sigma_{a,efective} = \frac{1}{\sqrt{2}} \sqrt{(\sigma_{1a} - \sigma_{2a})^2 + (\sigma_{2a} - \sigma_{3a})^2 + (\sigma_{3a} - \sigma_{1a})^2}$$

$$\sigma_{a,efective} = \frac{1}{\sqrt{2}} \sqrt{(153.54 - 76.77)^2 + (76.77 - 0)^2 + (0 - 153.54)^2} = 132.97 \text{ MPA}$$

The effective mean stress is:

$$\sigma_{m,efective} = \sigma_{1m} + \sigma_{2m} + \sigma_{3m} = 153.54 + 86.17 + 0 = 239.71 \text{ MPa}$$

And to obtain the completely reversible effective stress:

$$\sigma_{ar} = \frac{\sigma_{a,effective}}{1 - \frac{\sigma_{m,effective}}{\sigma_u}} = \frac{132.97}{1 - \frac{239.71}{476}} = 267.86 \text{ MPa}$$

From the material S-N curve we find the corresponded life to this stress:

$$N_f = \left[\frac{\sigma ar}{A}\right]^{\left(\frac{1}{B}\right)} = \left[\frac{267.86}{839}\right]^{\left(-\frac{1}{0.102}\right)}$$

$$N_f = 72651 \ \text{(cycles to failure)}$$

Calculated fatigue safety factor:

$$\frac{72651}{27375} = 2.65$$

2.65 > 2 (as requested)

Would the design still be safe if the internal fuselage pressure was to increase to 4.5 MPa?

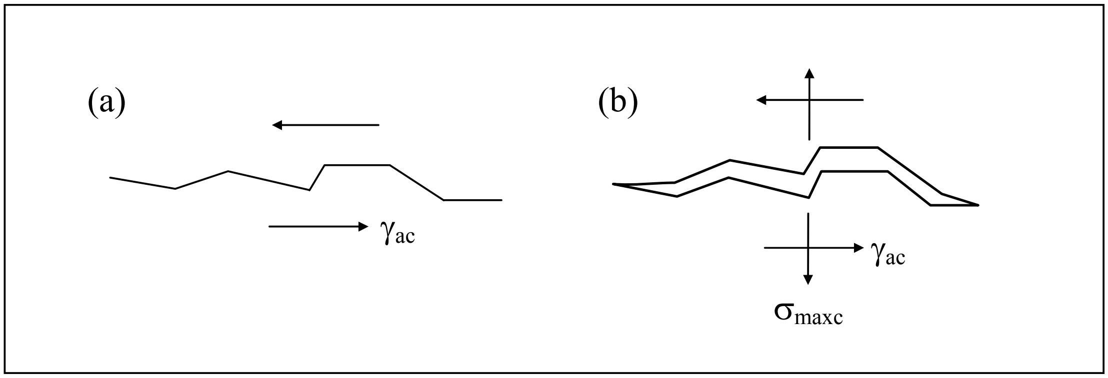
Example (in class exercise)

A solid shaft with 20mm diameter is subjected simultaneously to a constant (steady) bending of 200 Nm and alternating torsion of 220 Nm. Calculate the effective stress amplitude for this case if $\sigma_u$=450 MPa.
Hints: $S_{bending} = 32M/\pi d^3$ and $\tau_{torsion} = 16T/\pi d^3$ and note the torsion case.
(Answer: 560 MPa)

- **The critical plane approach**

The majority of the current and most accepted methods of fatigue assessments today apply the so-called **critical plane approach**. Some of these applications are very different and we only intend to introduce a few, well known, theories. The main difference between these methods and the classical approaches is in the understanding that the fatigue damage is related to the material micromechanical physical process.

The fatigue process is described in two stages: In the first stage slip bands' decohesion is primarily caused by shear and in the second stage a normal stress to the shear planes opens and augments the shear cracks. This process can be described schematically as shown below:




In figure (a) the crack is initiated on the plane of the maximum shear stress– fatigue stage 1. In Figure (b) the crack is open and grows due the normal stress to the plane of the maximum shear stress – fatigue stage 2. For high-strain fatigue case this process is modelled in terms of the maximum shear strain and the normal strain.

One of the main advantages of this approach is that any loading mode is already considered since the fatigue damage is accumulated directly on a critical plane in the component. However, the computation is fairly complex and only very recently have these models been introduced commercially.

**The McDiarmid model (stress approach)**

The McDiarmid parameter is based on finding a critical plane on which fatigue failure will occur. Fatigue strength is a function of the maximum shear stress and the normal stress occurring on the plane of the maximum shear stress. The critical plane for failure is that subjected to the greatest value of the parameter:

$$\boxed{\sigma_{equivalent} = \tau_{amp} + C_1\sigma_{n,max}}$$

where:

$\tau_{amp}$ = maximum shear stress amplitude
$\sigma_{n,max}$ = maximum normal stress occurring on the plane of maximum shear stress amplitude,
$C_1$ is a material constant

For cyclic conditions the McDiarmid parameter is therefore calculated using:

$$\boxed{\sigma_{equivalent} = \tau_{amp} + C_1\left(\sigma_{amp} + \sigma_{mean}\right)}$$

where:

$\sigma_{amp}$ = Stress amplitude normal to the plane
$\sigma_{mean}$ = Mean stress normal to the plane

The maximum value of shear stress $\tau_{max}$ is always at an angle 45° to the maximum principal stresses, and is calculated using:

$$\boxed{\tau_{max} = \frac{1}{2}(\sigma_1 - \sigma_3) \quad \text{where:} \quad \sigma_1 \rangle \sigma_2 \rangle \sigma_3}$$

On the plane of the maximum shear stress, the normal maximum stress is given by:

$$\boxed{\sigma_{n,max} = \frac{1}{2}(\sigma_1 + \sigma_3)}$$

**The Brown-Miller model (strain approach)**

In this theory it is assumed that failure is dominated by the maximum shear strain $\gamma_{max}$ and the strain $\varepsilon_n$ that acting normal to the maximum shear plane. The contribution of the normal strain is assumed to be governed by a material parameter (C) as follows:

$$\boxed{\frac{\gamma_{max}}{2} + C\,\varepsilon_n = \text{Const.}}$$

where: $$\boxed{\frac{\gamma_{max}}{2} = \frac{\varepsilon_1 - \varepsilon_3}{2}} \quad \text{and} \quad \boxed{\varepsilon_n = \frac{\varepsilon_1 + \varepsilon_3}{2}}$$

However, this theory does not correlate well all cases of multiaxial fatigue. It may be shown that under pure shear conditions the maximum shear strain plane is **on the surface** – this results with surface fatigue cracks. Under equibiaxial strain conditions the maximum shear strain plane is at $45^0$ through the thickness – results with **through thickness fatigue cracks**.

To improve the Brown-Miller fatigue model two cases of fatigue cracks are considered:

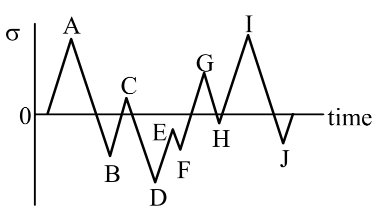
**TYPE A CRACKS** – SUFACE CRACK (max shear strain plane on surface)

**TYPE B CRACKS** – THROUGH THICKNESS CRACK (max shear plane inclined 45 degree to surface).

## Cycle Counting

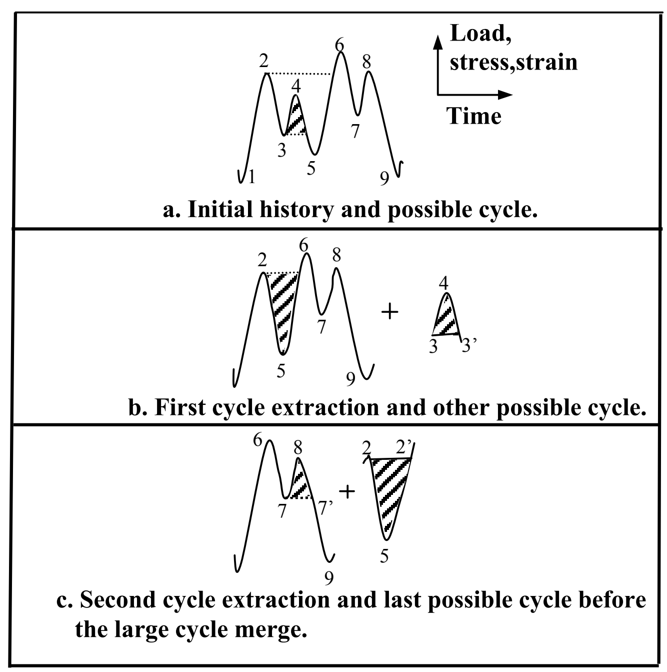
Service components in many cases experience variation of load with time, results with irregular stress history. It is therefore necessary to count individual events and define cycles that can be used for damage summation. Many debates concerning the summation of such events emerged in the past. Today, by far the most widely used procedure is called the **rainflow counting method**. Several standards have developed computer numerical algorithms for use of this method and an example of small sequence counting using the rainflow counting method is illustrated below:

## Definitions for irregular loading:




**Peaks: A, C**
**Valleys: B, D**
**Simple ranges:A-B, B-C**
**Overall ranges: A-D, D-G**

**Rainflow example:**



**Fatigue life prediction using finite element analysis (FEA)**

The use of elastic, static, finite element analysis (FEA) to estimate the critical cyclic stresses of service components is well established and has been widely used for the last 30 years. However, in many cases, at the critical location, the local stress is above the elastic limit and in order to establish the cyclic stress or strain or the critical hysteresis loop, a non-linear elastic-plastic FEA is required. The principles of fatigue estimate using an elastic-plastic FEA are briefly described and an example is shown for estimation of life of a component

Typical elastic-plastic analysis required the following material input data:

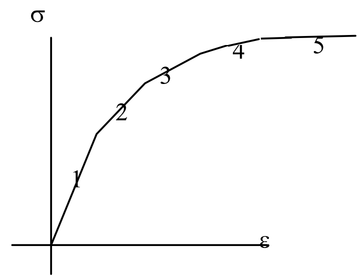
1. **Elastic-plastic cyclic stress-strain response.**
2. **Hardening low or the elastic-plastic material response upon unloading.**
3. **Fatigue strain-life low such as the Manson-Coffin Basquin relationship.**
4. **Multiaxial fatigue criteria such as the Brown-Miller parameter.**

For the elastic-plastic FEA a non-linear analysis is performed and iterative solution procedure is required using convergence criteria. Most modern FEA codes have internal convergence criteria and the analysis iterates automatically until the convergence criterion is satisfied.

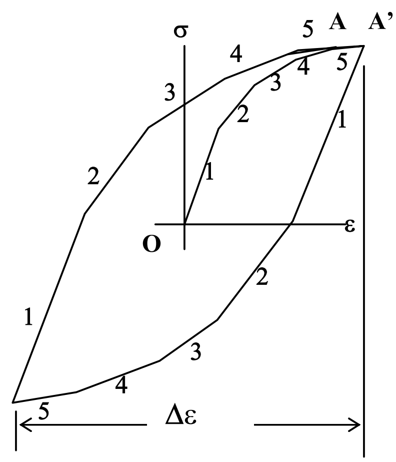
Usually the loading history is sub-divided into a certain number of load increments to improve the convergence of the non-linear iterative procedure.

A typical elastic-plastic fatigue FEA procedure is illustrated overleaf, the material cyclic stress-strain response is shown in (1), and the stress-stain cyclic loop resulting from loading O-A-B-A' in (3) is shown in (2). The simulated strain results are used as input for a strain based durability analysis that assumes that the life ($N_f$) of the component is dependent on the strain range ($\Delta\varepsilon$) in a critical location (4). It is normally sufficient to model one half of the strain range cycle by using the segment O-A in (3).

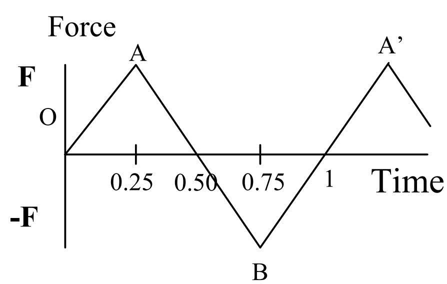
### FEA durability (fatigue) procedure – constant amplitude load:



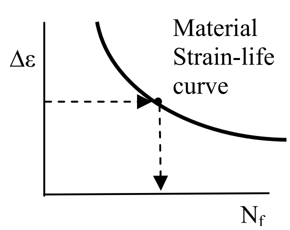

(1) The material cyclic response is entered as a stepwise value of stress-strain or as a non-linear equation.




(2) The material hardening model, for example, kinematics or isotropic behaviour is selected to describe the elastic-plastic unloading part of the hysteresis loop.



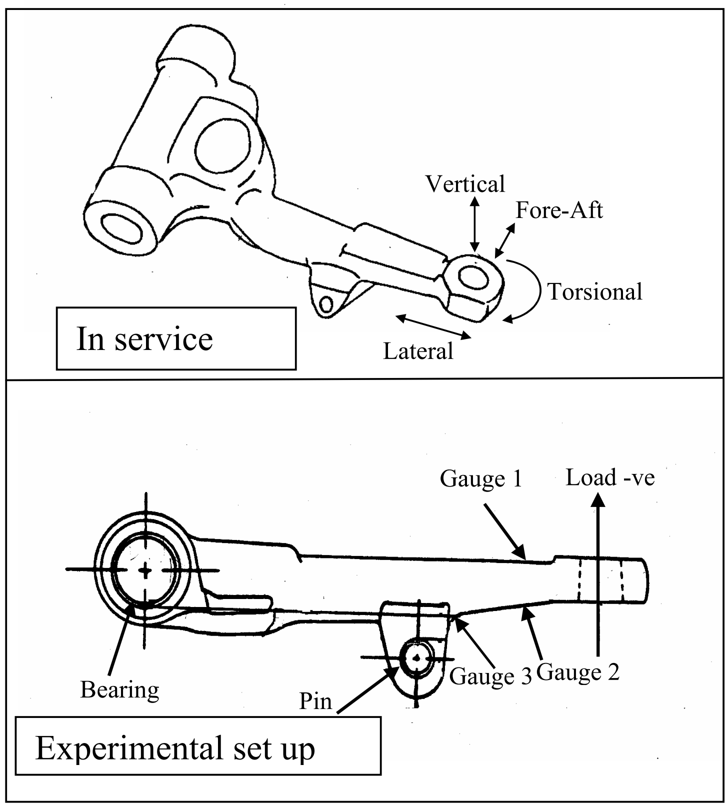

(3) Incremental loading is entered for a complete cycle.




(4) Estimation of the component life using the FEA results and strain-life curve.

**Example 2**

Several years ago Rover design engineers investigated premature failure of the suspension arm in some of the Metro cars front wheels (fiction only). The load pattern applied to this component is complex and includes 3 directions of direct load and at least one component of torsional load – see illustration. An experimental programme was carried out using the service attachments with a simplified load – only in the vertical direction. The component was strain gauged and fatigue cycled to failure.
The laboratory load was higher than that used in service and elastic-plastic durability FEA analysis was requested to confirm the experiments.

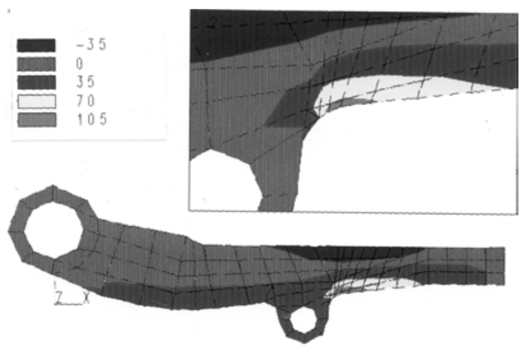



The solution stages used are shown below:

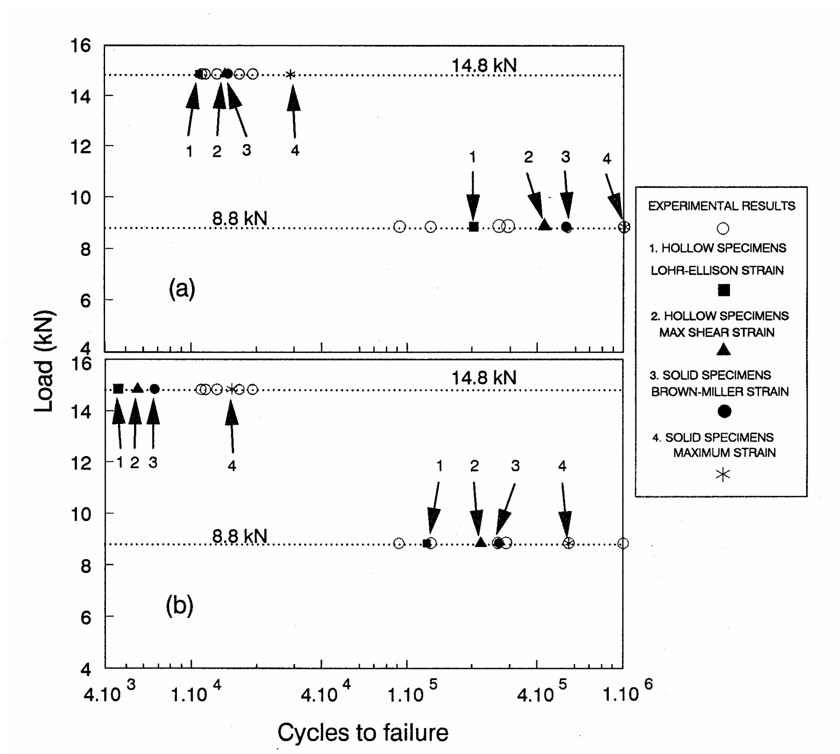
Two FEA models were used; assuming plane stress condition or assuming plane strain condition. The stages described previously were applied using a commercial FEA code.

Initially, strain gauge results obtained from static elastic tests were compared to the FEA analysis. The results are shown below for the plane-stress conditions.

An example of the FEA analysis results:




1. Results from a simplified model using static plane stress analysis.




2. Comparison between FEA results and the experimental results using several fatigue life prediction models (Shatil et al, 1994).

**Advanced Fatigue Tutorials**

1. A cylindrical pressure vessel with closed ends is made from the alloy Ti-6Al-4V (A=1889, B=-0.104, $\sigma_u$ = 1233 MPa) and has a diameter of 250mm and a wall thickness of 2.5mm.
   a. What applied pressure will cause fatigue failure at $10^5$ cycles? (14.6)
   b. What is the maximum pressure applied if the vessel is designed for endurance (assume factor of safety of 2)? (6.5)
   c. Calculate the minimum cylinder thickness, designed for endurance as in b. and using the pressure obtained at a.

2. In a paper by E K Walker, it is proposed that mean stress effects can be estimated by using: $\sigma_{equivalent} = \sigma_{max}(1\text{-}R)^m$ where $\sigma_{equivalent}$ is an equivalent stress range at R=0 that causes the same fatigue failure as the actual $\sigma_{max}$ for any given mean stress, and m is a material constant.

   a. Modify the above equations so it is expressed in terms of equivalent completely reversed (R= -1) stress amplitude, $\sigma_{ar}$.
   b. Based on the above develop an equation for the particular case when m=0.5.

3. a) Develop expression for the elastic-plastic **uniaxial** case of the Brown-Miller model if $\varepsilon_1$, the maximum principal strain is known and assuming the following material properties: C = 0.2, $\nu_{elastic}$ = 0.3 and $\nu_{plastic}$ = 0.5. Note also that for the uniaxial case $\varepsilon_3 = \varepsilon_2 = -\nu\varepsilon_1$.
   b) Modify the strain life equation to consider the Brown-Miller model in the uniaxial case.

# Introduction to LEFM

**Design for damage tolerance**

It is generally accepted that most service components or structures contain some form of defect or flaw by the time manufacturing is completed. Fracture control of a structure is the concerted effort by the design engineers to guarantee safe operation by ensuring that any crack developed from the flaws is 'sustained' during the service load history.

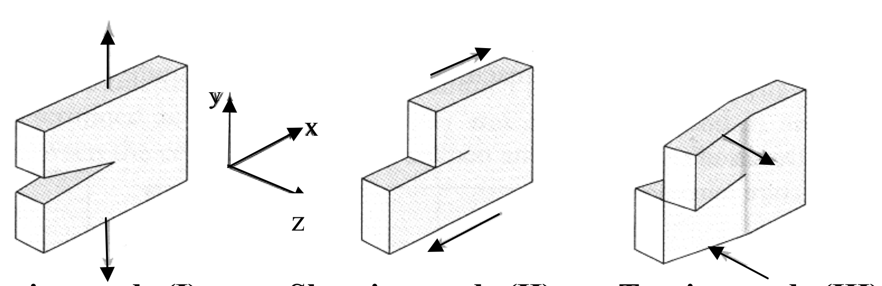
Fracture control is the combination of measures to prevent in service fracture and includes the damage tolerance analysis, material selection, design improvements, experimental programme and maintenance inspection and replacement schedule.

The Damage Tolerance analysis has two main objectives:
1. Predict the effect of cracks on the structural strength (fracture margins).
2. Predict crack growth during service history (fracture control).

Practical fracture design commonly distinguishes between two types of fracture mechanisms:
1. **Brittle fracture** is that associated with cleavage fracture and results with 'sparkling' differently orientated fractured surfaces.
2. **Ductile fracture** is that associated with the initiation of voids from inclusions and results with dull looking fracture surfaces.

During ductile fracture some local plastic deformation always occurs next to the crack tip.

In fracture analysis the applied stress (far field stress) is related to the processes taking place at the crack tip. Three modes of fracture are distinguished as shown below. However, in practical applications Mode 1 fracture is usually the most important and our analysis relates to Mode 1 fracture.




**Opening mode (I)** **Shearing mode (II)** **Tearing mode (III)**

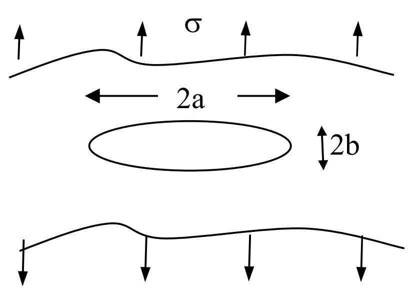
**Modes of fracture**

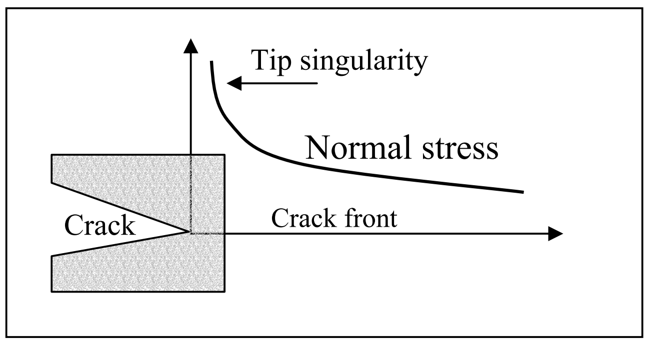
**The stress intensity factor K**

The principles of linear elastic fracture mechanics (LEFM) are based on work by Griffith and Irwin. Griffith has demonstrated that for a narrow ellipse in a wide plate, where the stress concentration is define as:

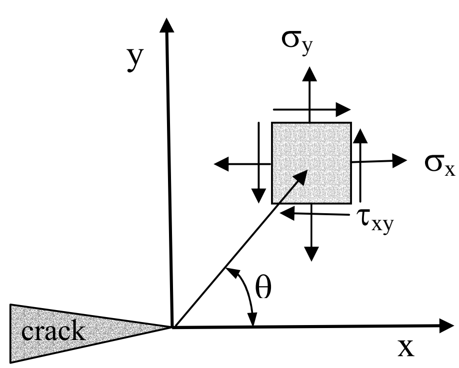
$$\boxed{K_t = 1+(2a/2b)}$$

as the ellipse becomes narrower, or a narrow slot (resembling a crack) b → 0 the normal stress increases to infinity or Kt → ∞. This is shown schematically below.




This approach was further developed to the practical design of fracture in metals by Irvin using the equations for ealstic stress distribution at the crack tip for arbitrary stress and fracture modes as follows:



And for Mode 1 only the crak tip stress field is a dependent on r and θ as follows:

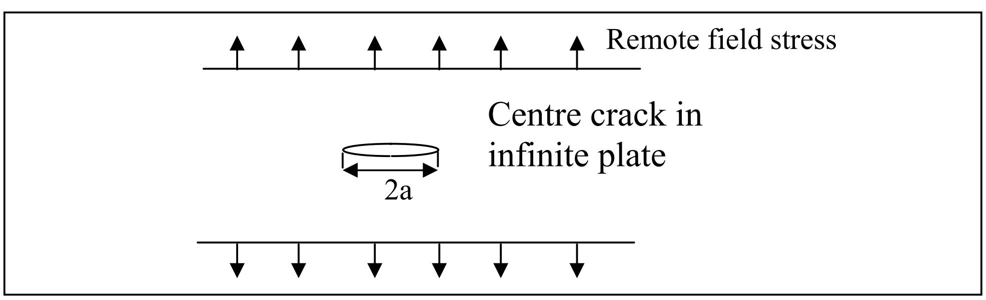
$$\sigma_x = \frac{K_1}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\left[1 - \sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right]$$

$$\sigma_y = \frac{K_1}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\left[1 + \sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right]$$

$$\sigma_z = 0 \qquad \text{(plane stress)}$$
$$\sigma_z = \upsilon(\sigma_x + \sigma_y) \qquad \text{(plane strain)}$$

$$\tau_{xy} = \frac{K_1}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\sin\frac{\theta}{2}\cos\frac{3\theta}{2}$$

For the particular case of a centre crack in infinite plate below, it can be shown that the stress normal to the crack tip, $\sigma_y$, is defined by the following:




$$\boxed{\sigma_y = \frac{C\sigma\sqrt{a}}{\sqrt{2\pi x}}}$$

And, for a crack through a plane with θ = 0 the stress at the crack tip is reduced to:

$$\boxed{\sigma_y = \frac{K}{\sqrt{2\pi x}}}$$

Comparing the two equations and assuming that: $C = \sqrt{\pi}$
(refer to literature for full solution)

We can define now the basic equation for the **stress intensity factor, K:**

$$\boxed{K = \beta\sigma\sqrt{\pi a}}$$

Where a is half crack length as defined above, σ is the remote field stress and β is a geometrical factor.

For infinite plate above: $\boxed{\beta = 1}$

For finite plate (or use the Table below) : $\boxed{\beta = \sqrt{\sec\frac{\pi a}{W}}}$

Since the equations developed above relate to Mode 1 fracture the stress intensity factor is denoted as:

$$\boxed{\mathbf{K = K_1}}$$

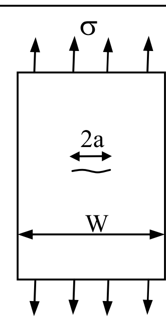
β solutions for three common crack geometries are given below:

**Centre crack in finite width plate:**

$$\beta = 1 + 0.256\frac{a}{W} - 1.152\left(\frac{a}{W}\right)^2 + 12.20\left(\frac{a}{W}\right)^3$$

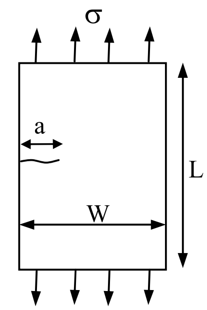
or

$$\beta = \sqrt{\sec\frac{\pi a}{W}}$$



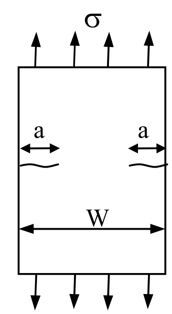
**Single edge crack in finite width plate:**
**L/W = 2 only:**

$$\beta = 1.12 + 0.23\frac{a}{W} + 10.56\left(\frac{a}{W}\right)^2 - 21.74\left(\frac{a}{W}\right)^3 + 30.42\left(\frac{a}{W}\right)^4$$




**Double edge crack in finite width plate:**

$$\beta = 1.12 + 0.43\frac{a}{W} - 4.79\left(\frac{a}{W}\right)^2 + 15.46\left(\frac{a}{W}\right)^3$$




**Notes:**

The stress intensity factor K is a means of characterising the elastic stress distribution near the crack tip but it has no physical meaning!

The stress intensity factor, K, has the units of:
$MNm^{-3/2}$ **or** $MPa\sqrt{m}$ **or** $N/mm^{3/2}$ (all same SI units!).

**The fracture toughness $K_{1c}$**

One of the most important design implications of LEFM is the use of the fracture toughness, $K_{1c}$. Since K defines the entire stress field near the crack tip, fracture occurs when it reaches a critical value. **The value of the critical K is a material dependent property** and is determined from established experimental procedures.

The fracture toughness is always determined for **PLANE-STRAIN** condition only. If an assumed or detected defect exists in a structure, the designed stress must not exceed a critical fracture stress for a given material. For example, for a crack at an infinite plate:

$$\boxed{\sigma_{fracture} = \frac{K_{1c}}{\sqrt{\pi a}}}$$

Typical fracture toughness values for some common materials:

| **Material** | $K_{1C}$ ($MPa\sqrt{m}$) | $G_{1C}$ ($kJ/m^2$) |
|---|---|---|
| Mild steel | 150 | 70 |
| HSS | 90 | 58 |
| Cast iron | 13 | 1.6 |
| Titanium alloys | 75 | 65 |
| Aluminium alloys | 34 | 19 |
| Glass | 0.5 | 0.02 |
| Epoxy | 0.6 | 0.2 |

**Trends of $K_{1c}$**

- Metals typically have toughness values in the range of 20 to 200 $MPa\sqrt{m}$.
- Polymers toughness values are in the range of 1 to 5 $MPa\sqrt{m}$
- Ceramics toughness values are about 1 to 10 $MPa\sqrt{m}$.
- Increase in metal strength DECREASES the fracture toughness.
- Heat treatment of metals DECREASES the fracture toughness.
- Loading rate INCREASES toughness of steel

**IN GENERAL: INCREASE DUCTILITY = INCREASE in $K_{1c}$**

**Another Note on $K_{1c}$**

The material $K_{1c}$ value is obtained under carefully controlled plane-strain test condition and therefore it is the most conservative value of the critical fracture for a particular material. This value is used for design against catastrophic failure and is considered to be dependent on the material only.

**Example**

Determine the maximum load that the component shown below can withstand if it is made of Ti 6Al 4V ($\sigma_y = 900$ MPa, $K_{1c} = 100$ MPa m$^{1/2}$).

Solution:

a = 10mm  W = 100mm  a/W = 0.1

For a single edge notch using the geometrical solutions above:

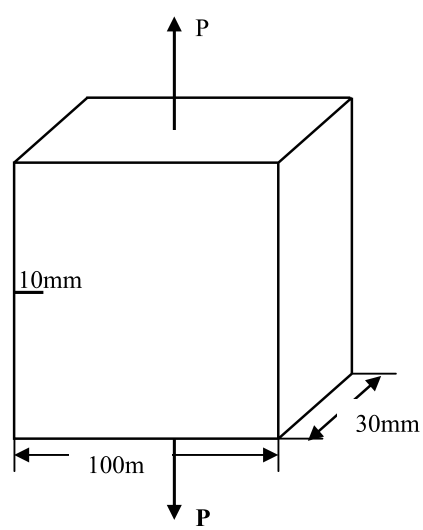
$$\beta = 1.12 + 0.23x0.1 + 10.56(0.1)^2 - 21.74(0.1)^3 + 30.42(0.1)^4 = 1.23$$

Using the fracture toughness to obtain the critical stress:

$$\sigma_{\max imum} = \frac{K_{1c}}{\beta\sqrt{\pi a}} = \frac{100}{1.23\sqrt{\pi 0.01}} = 458.7 \text{ MPa}$$

Since 458.7 < 900 the component fracture before yield occurs.

$P_{max} = \sigma A = [458.7x10^6]$ x [0.1 x 0.03] = 1.376 MN

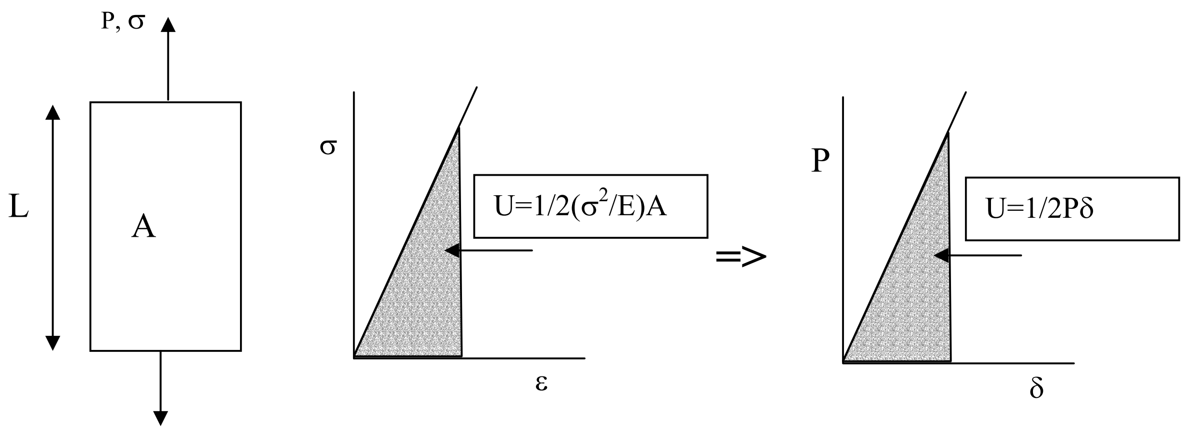


**Energy methods for fracture**

Historically, the energy methods superseded the development of the intensity factor, in particular, **Griffith and Irwin models** that are commonly called:

**The Strain Energy Release rate, G**

In the case of a simple tensile bar the stress is uniform and the total strain energy of a member with a cross section area A and length L is simply the area under the elastic stress-strain diagram as shown below, summed up over the entire member volume:




Since: $\delta = \varepsilon L = \sigma L/E$, and $P = \sigma A$ for the elastic case shown.

The energy release rate due to the extension of a crack is shown below. The total balance to keep the energy conservation criterion can be written as the positive change in external energy due to the cracking, dF/da that is equal to the internal or strain energy, dU/da and the fracture energy dW/da, thus, the fracture energy is:

$$\boxed{\frac{dW}{da} = \frac{d}{da}(F - U)}$$

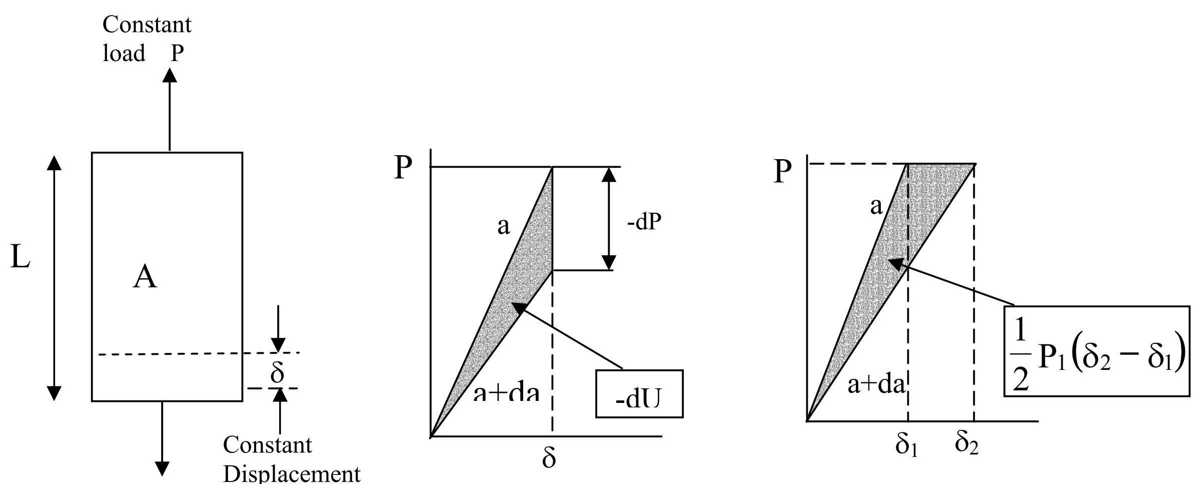
**Two cases must be considered**:

1. The crack is increased by $\Delta a$ but the displacement is constant (Figure A below). In this case the load is actually dropping so no additional external work is required. Therefore dF/da = 0 and the total energy balance is negative:

$$\boxed{\frac{dW}{da} = \frac{d}{da}(-U)}$$

2. The crack is increased by $\Delta a$ but the load is constant and the change is in displacement ($\delta_2$ - $\delta_1$), Figure B below. The load $P_1$ is doing the work equal to the change in the strain energy:

$$\boxed{\frac{dW}{da} = \frac{1}{2}P_1(\delta_2 - \delta_1) = \frac{dU}{da}}$$




**A** – Constant displacement

**B** – Constant load

From this we conclude that the change in fracture energy is always equal to the change in the strain energy, regardless to the loading conditions or the numerical sign!

OR: $\boxed{\dfrac{dW}{da} = \dfrac{dU}{da}}$ $\longrightarrow$

**$G = R$ where:**

**$dU/da$ = $G$ = Energy release rate**
**$dW/da$ = $R$ = Fracture resistance**

It can be shown that fracture occurs when:

$$\boxed{R = \frac{dW}{da} = \frac{\pi \sigma^2 a}{E}}$$

Comparing this expression with the expression for the stress intensity factor:

$$\boxed{K = \sigma\sqrt{\pi a}}$$

We obtain an expression for fracture due to energy calculations:

$$\boxed{K = \sqrt{ER}}$$

OR, alternatively the **fracture resistance** due to a critical **toughness** is:

$$\boxed{R = \frac{K_{1c}^2}{E}}$$

The critical fracture energy release rate $G_C = R$

## Plane stress Vs Plane strain:

The quantities of R and K can be related to the geometry as follows:

$$\boxed{R = \frac{K_{1c}^2}{E'}}$$

Where

| | | | |
|---|---|---|---|
| E’ | = | $E$ | (plane stress, $\sigma_z = 0$) |
| E’ = | | $\dfrac{E}{1-\upsilon^2}$ | (plane strain, $\varepsilon_z = 0$) |

**Example (in class):**

**a.** A steel plate 750mm wide and 5mm thick with a central crack 2a = 100mm, is pulled to fracture. Fracture occurs at 800 kN. What is the stress intensity at fracture?

For this geometry assume $\beta \cong 1$.

**b.** Use fracture energy considerations to calculate the fracture stress of the same plate made of the same steel as above (E = 205 GPa) but with an edge crack length of a = 75mm.

The fracture energy for this material is:

$R = K^2/E = (84.55)^2/205x10^3 = 0.03487$ [MPa m] = 34870 N/m

and for the new plate at fracture:

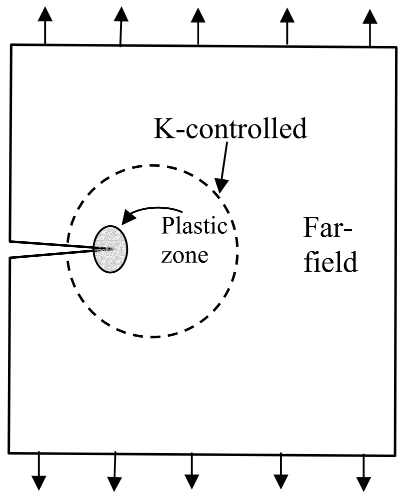
$\beta = 1.12$

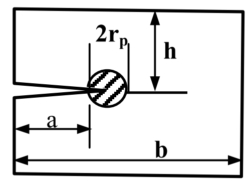
$$R = \frac{(\sigma_{fr}\beta)^2 \pi a}{E}$$ therefore:

$$\sigma_{fr} = \sqrt{\frac{ER}{\beta^2 \pi a}} = \sqrt{\frac{205000 x 0.03487}{1.12^2 x \pi x 0.075}} = 155.5 \text{ MPa}$$

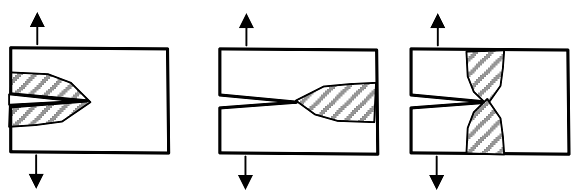
## Crack Tip plasticity corrections (Local yield)

The main limitation of LEFM is that it assumes a very localised yield near the crack tip. This is demonstrated in the figures below:




Valid LEFM



Invalidated LEFM due to plastic zone too large

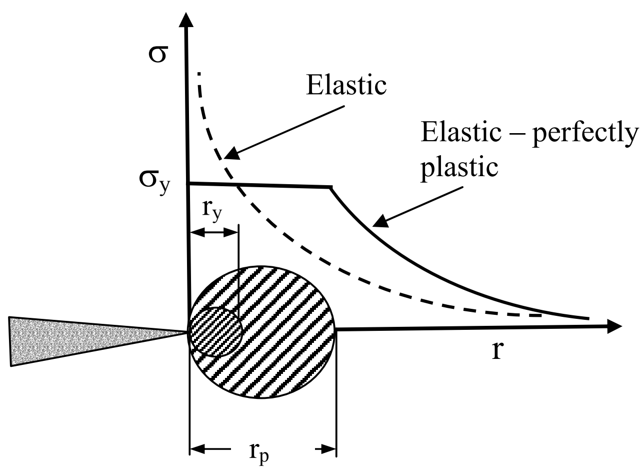
LEFM is applicable only if the limitations on (a) (crack length) (b-a), (ligament length) and (h) (edge distance) are satisfied. This is expressed as:

$$\boxed{a, (b-a), h \geq \frac{4}{\pi}\left(\frac{K}{\sigma_y}\right)^2 \qquad \text{(LEFM} \quad \text{applicable)}}$$

## Irwin plastic zone size

The classical LEFM approach assumes singularity at the crack tip. The normal load is asymptotic to the y-axis and goes to infinity, as the distance is closer to the crack tip, as shown below. However, due to the material yield this does not occurs in **ductile materials** and the stress follows the material stress-strain response.

To estimate the radius of the plastic zone size from the crack tip we assume a perfectly plastic material and the material near the crack tip follows the modified stress distribution as shown below.




## Plane stress correction

From the previous LEFM stress field solution it can be shown that the plastic zone size radius, $r_y$, for plane stress conditions is as follows:

$$\boxed{r_y = \frac{1}{4\pi}\left(\frac{K_1}{\sigma_y}\right)^2}$$

However, this assumes elastic stress distribution and from consideration of elastic-perfectly-plastic redistribution of the stresses it is shown that the plastic zone size is twice as large (plane-stress):

$$r_p = 2\,r_y = \frac{1}{2\pi}\left(\frac{K_1}{\sigma_y}\right)^2 \quad \text{plane} - \text{stress}$$

**Plane strain correction**

Similar analysis is carried out for a plane strain condition. In this case the plastic zone is constrained by the triaxial stress state. For the perfectly-plastic material this is shown to be about one third the size of that under plane stress condition:

$$r_p = 2\,r_y = \frac{1}{6\pi}\left(\frac{K_1}{\sigma_y}\right)^2 \quad \text{plane} - \text{strain}$$

Since the redistribution of the stresses near the crack tip is higher than that under LEFM assumption an **effective stress intensity** factor is calculated using an **effective crack length:**

$$a_{eff} = a + r_p$$

An effective stress intensity factor is computed using the effective crack length:

$$K_{eff} = \beta_{\text{function of } a_{eff}}\, \sigma \sqrt{\pi\, a_{eff}}$$

Since the effective crack size, $a_{eff}$, is taken into account in the geometry correction, $\beta$, an iterative solution is usually required to solve $K_{eff}$. The K is initially determined assuming LEFM solution. A first estimate of $a_{eff}$ from the equation above is used to compute a new $K_{eff}$. From this new $K_{eff}$ a new $a_{eff}$ is calculated. This process is continued until consecutive values of $K_{eff}$ converge. Typically, only 3-4 iterations are required.

**Limitations of LEFM**

The analysis carried out in this section assumed that the crack tip behaviour is described by what is commonly called the '*region of K-dominance*', even if it is beyond the region of the classic LEFM stress field.

However, if this no longer applies, the K-analysis can no longer be used and other methods apply using **ELASTIC-PLASTIC FRACTURE MECHANICS** parameters and theory. As a 'rule of the thumb' the stress applied must be below 80% of the fully plastic material yield.

**Example (in class exercise LEFM-1)**

A high-strength steel plate with a through thickness center crack length of 2a = 20mm is subjected to tension stress of 400 MPa. If the yield strength of this steel is 1500 MPa, (assume $\beta = 1$):
a) what is the plastic zone size and the stress intensity factor for the crack? Assume plane stress and infinitely wide plate.
b) What effect does the plastic zone has if the steel used has a yield strength of 400 MPa?
c) Assume plane-strain and yield stress of 400 MPa and recalculate.

**Example (in class exercise LEFM-2)**

A linkage in the landing gear of a jet plane is to be fabricated from a HSS, which could be either:
(I) An alloy steel $\sigma_y = 1900$ MPa, $K_{1c} = 82$ $\text{MPa}\sqrt{\text{m}}$
(II) Medium carbon steel $\sigma_y = 1000$ MPa, $K_{1c} = 50$ $\text{MPa}\sqrt{\text{m}}$

Which of these two steels has the better tolerance to defects in terms of toughness and critical crack length? (Assume factor of safety of 2). How can you achieve the same structural damage tolerance for both materials?

**Example (in class exercise LEFM-3)**

During routine inspection of commercial aircraft, a 2.5mm edge crack was detected in a 50mm wide and 100mm long panel that is part of the wing leading edge. The crack direction was approximately normal to the operating tension load. The panel is made of an aircraft graded aluminium alloy 7075-T651 ($\sigma_y$ = 505 MPa, $K_{1c}$ = 29 MPa $\sqrt{m}$). Assume factor of safety of 2.

1. Estimate whether the panel is safe in terms of critical fracture assuming that the maximum allowable stress is applied.
2. What will happened if the crack grows another 1mm?
3. Estimate the critical crack length (some iteration required).

Solution procedure:

1.a Determine type of crack and calculate stress intensity factor.
1.b Compare to the material toughness property.
2. Add 1mm and repeat 1a and 1b above.
3. Compare stress intensity for several values of crack length. Try to converge so that a particular crack length yields the required material fracture toughness.

(Solution will be given during class)

**Solution (LEFM-1)**

a. For plane stress infinite plate ($\beta$ = 1) the plasticity zone size is:

$$r_p = \frac{1}{2\pi}\left(\frac{K_1}{\sigma_y}\right)^2 = \frac{1}{2}\left[\frac{\sigma^2 a}{\sigma_y{}^2}\right] = \frac{1}{2}\left[\frac{400^2 \text{ x } 10}{1500^2}\right] = 0.356 \text{ mm}$$

and therefore the effective stress intensity is:

$$K_{eff} = \sigma\sqrt{\pi a_{eff}} \qquad \text{where} \qquad aeff = a + rp$$

$$K_{eff} = 400\sqrt{\pi \text{x} (10 + 0.356) \text{x} 10^{-3}} \quad = \quad 72.15 \quad \text{MPa}\sqrt{m}$$

However, rearranging the plasticity correction equations we obtain a direct relationship based on crack length and the applied stress:

$$K_{eff} = \sigma\sqrt{\pi a_{eff}} = \sigma\sqrt{\pi}\sqrt{a + \frac{1}{2\pi}\frac{\sigma^2 a}{\sigma_y{}^2}} = \sigma\sqrt{\pi a}\sqrt{1 + \frac{\sigma^2}{2\sigma_y{}^2}}$$

$$K_{eff} = 400\sqrt{\pi \,x\, 0.01}\sqrt{1 + \frac{400^2}{2 \,x\, 1500^2}} = 400 \,x\, 0.1772 x \sqrt{(1 + 0.0356)} = 72.13 \text{ MPa}\sqrt{m}$$

b) Answer : 86.8 MPa $\sqrt{m}$ c) Answer : 76.7 MPa $\sqrt{m}$

**Solution (LEFM-2)**

To estimate the crack length at critical fracture we use the fracture toughness definition to obtain critical defect for each steel:

For the **alloy steel**:

The maximum allowable stress is: $\sigma_{allowable} = \sigma_y/2$ = 1900/2 =950 MPa.

Using the Fracture toughness equation:

$$a_{critical} = [illegible]$$

For the **carbon steel**:

$$a_{critical} = [illegible]$$

The critical crack length is 2a, so it is 4.8mm for the first and 6.36mm for the second material.

It shows that the medium-carbon steel is preferred in terms of damage tolerance!

To get a toughness or stress intensity K for which the alloy steel will have the same damage tolerance:

Alternatively we can calculate a new allowable stress:

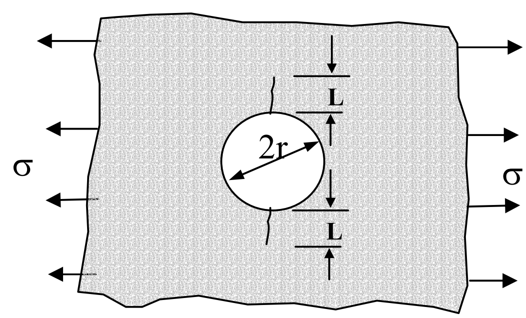
Or a larger factor of safety of:

$$1900/820 = 2.3$$

# Tutorials

1. A large sheet of aluminium alloy is loaded in tension to a stress of 200 MPa. If its yield stress is 400 MPa and fracture toughness is 42 MPa $m^{1/2}$, what is the maximum tolerable defect size (i) Using LEFM (ii) using plasticity zone correction? (14mm, 12.3mm)

2. A large plate carrying a tensile load contains a hole with two cracks propagating from the hole in a direction perpendicular to the applied load, see figure below. (Assume that $\beta = 1.12$ for edge crack plate and $\beta = 1$ for infinite plate).




   (a) Set up the stress equation at which the plate will fracture by both of the following methods.
      (i) Assume the stress intensity in the cracks is the same as the stress intensity in the edge of an infinite plate whose average stress is the same as the peak stress round a hole in an uncracked plate.
      (ii) Assume that the only effect of the hole is to make the two cracks at opposite sides of the hole appear as one long continuous crack.
   (b) At what ratio r/L do both approaches give the same estimate? Which approach is more likely to be correct for r « L?

3. The stress intensity for a partial-through thickness flaw is given by $K = \sigma(\pi a)^{1/2}(\sec(\pi a/2t))^{1/2}$ where a is the depth of penetration of the flaw through a wall thickness t. A flaw is 5mm deep in a wall 12mm thick made from 7075-T6 aluminium alloy ($K_{1c}$ = 24 MPa $m^{1/2}$) that support a stress of 172 MPa. What will be the critical stress level for this structure and what is the maximum crack length this structure can withstand? (Some iterations may be required). (Answer: $\sigma_{max}$ = 171 MPa, a = 4.9mm)

4. A ship steel has a value of $G_c$ = 35 kJ m$^{-2}$ and E = 205 GPa.
   (a) What is the fracture stress in a thin plate that is 300 mm wide and that contains a central crack 12 mm long? (617 MPa)
   (b) If the crack is 50 mm long, what is the fracture stress? (214 MPa)
   (c) Increasing the plate thickness to 120 mm reduces $G_c$ to 18 kJ m-2. What is the fracture stress for a 12 mm-long crack? (442 MPa)

5. A cylindrical pressure vessel 8m long 4m in diameter exploded at an internal pressure of 10 MPa. The vessel is made from 20mm thick steel with the following properties: E = 210 GPa; $\sigma_y$ = 1.2 GPa; $G_c$ = 120 kJ m$^{-2}$. Show that the failure is compatible with the existence of a 16mm longitudinal crack. Assume that 2c=2a where 2c is the surface crack length and 2a is the flaw depth.

6. A cylindrical steel pressure vessel with yield strength of 360 MPa is subjected to a hoop stress of 140 MPa. A tensile residual stress of 80 MPa can also be assumed to be present.
   The lowest service temperature of the vessel will be -25°C, under which conditions the $K_{1c}$ is 44 MPa m$^{1/2}$. The vessel will be designed according to a "leak-before-break" philosophy in which a detectable leak will occur before brittle fracture could occur. Determine the thickness of the pressure vessel based on fracture mechanics considerations. Assume 2a=2c as above. (10.5mm)

# Fatigue Crack Growth

**Introduction**

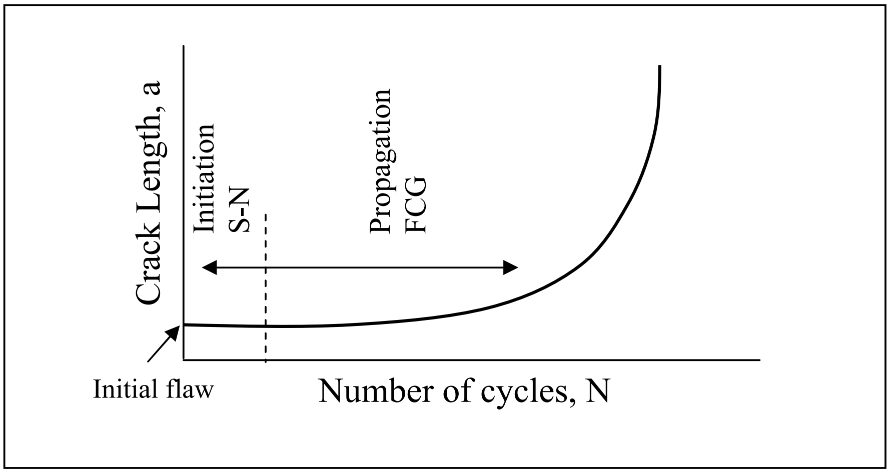
- The material in the preceding lecture has dealt mainly with static or monotonic loading. We now consider crack growth in the presence of cyclic stresses.

- About 40 years ago a linear relationships was shown to exits for most **ductile materials** between the stress intensity and the rate of crack growth. Since that times the application of fracture become an integral part of fatigue analysis in many application of structural design.

- In our previous fatigue lectures we have assumed that no initial defect exits in the component and therefore the analysis was solely to establish the number of cycles to **crack initiation**. However, most structure could be designed to withstand a certain level of crack growth assuming that a manufacturing defect exists in the part.

- The same limitations of LEFM also govern the FCG analysis, this restrict the crack tip stress field so it is dominated by elasticity considerations. This implies that it is mainly in the HCF regime.

- A typical modern fatigue analysis may initially consists of calculation of the number of cycles to crack initiation using the S-N diagram and in the second stage apply FCG analysis to evaluate the number of cycles up to a critical crack length.



**Principles of FCG**

For constant amplitude cyclic loading the range and ratio of the nominal stresses were previously described as:

$$\Delta S = S_{max} - S_{min} \quad \text{and} \quad R = S_{min}/S_{max}$$

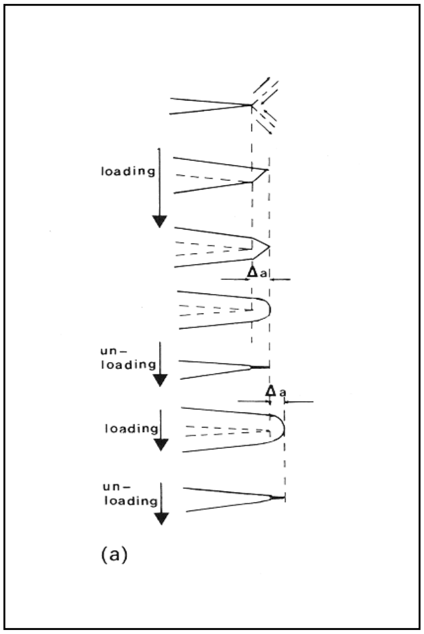
where $S_{max}$ and $S_{min}$ are the maximum and minimum nominal applied stresses.

Using the LEFM definitions for stress intensity we obtain:

$$K_{max} = \beta S_{max}\sqrt{\pi a} \qquad K_{min} = \beta S_{min}\sqrt{\pi a}$$

$$\Delta K = \beta \Delta S \sqrt{\pi a} \qquad R = \frac{K_{min}}{K_{max}}$$

The fatigue process at the crack tip could be illustrated as follow:

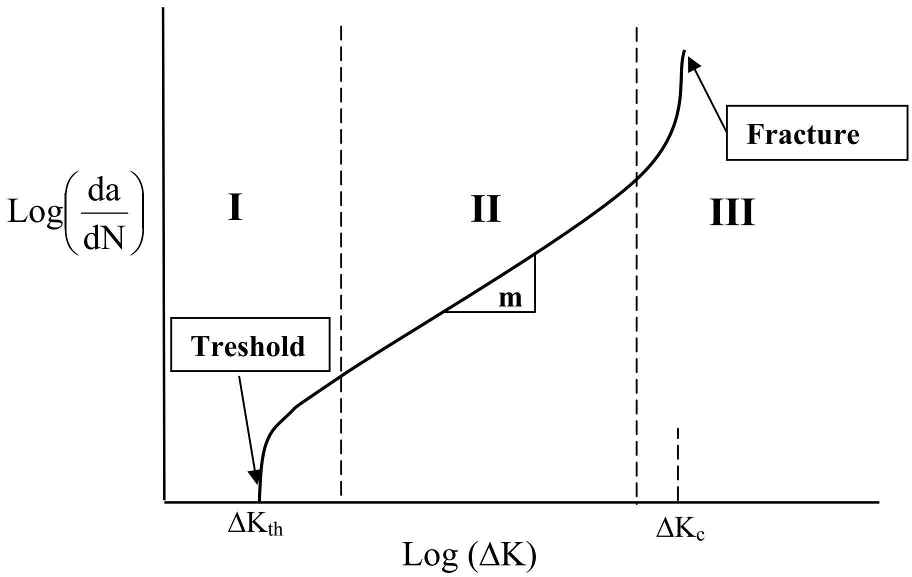



1. PSB
2. Coalescent
3. Propagation
4. Blunting
5. Closure
6. Repeat

**The Paris relationships**

At the basics of FCG analysis is the fundamental relationship between the stress intensity range, **ΔK** and the **CRACK GROWTH RATE da/dN** (or how much the crack propagates per number of cycles).

This is plotted on a log-log diagram as shown below with the rate of crack propagation, da/dN on the abscissa and the range of stress intensity, ΔK on the x axes as shown. The curve below, demonstrate the FCG behaviour in metals, has typically three distinct regions:

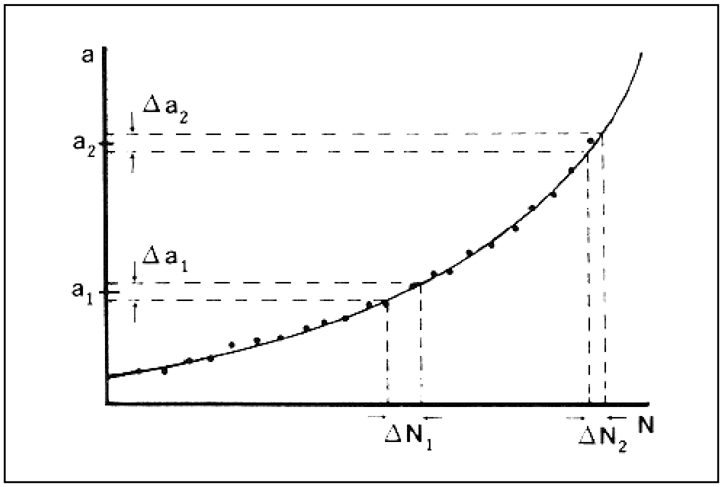



(I) Slow growth near the threshold value, $\Delta K_{th}$**.**
(II) Intermediate region of stable growth follows a power law.
(III) Unstable rapid growth up to a critical value, $\Delta K_{cr}$.

The equation that represent the **stable growth** (region B) is identified with **Erdogen** and **P C Paris** and is written as a power law:

$$\boxed{\frac{da}{dN} = C(\Delta K)^m}$$

The crack growth rate per cycle is measured in a well defined experiments that are specified in standards such as the BS and ASTM and the data obtained as shown below.




**Effects of R-ratio on the FCG**

It has been shown that the Paris FCG equation above is dependent on the R-ratio, or the ratio between maximum and minimum applied stresses, similar to the S-N curve, and a different curve is generated for each R ratio. In many cases the curves for different R have similar slope (for the same material). This means that they have the same **m** values but different **C** values in the Paris FCG equation above.

The basic material FCG relation is usually obtained for **R = 0** where no compression or crack closure involved in the fracture process. A large data base have been accumulated for many structural materials of FCG curves at different R ratios and it is possible, in many cases, using ESDU data for example, to interpolate from those curves a crack growth rate for a particular stress intensity range at a particular R.

Alternatively, various empirical modifications of the FCG equation were introduced that remove the dependency on the R ratio. Two of the most widely used are the Walker and Forman modifications shown below.

**Walker correction:**

$$\boxed{\frac{da}{dN} = C\left[\frac{\Delta K}{(1-R)^{1-\gamma}}\right]^m}$$

Where c and m are material constants at R =0

where C and m are the FCG constants for $\Delta K = 0$ curve and $\gamma$ is a material property, typically varies from 0.3 to 1. Note also that this equation gives a family of parallel straight lines, e.g. assume a constant slope, m.

**Forman Correction**

$$\frac{da}{dN} = \frac{C(\Delta K)^m}{(1-R)(K_{1c}-K_{max})}$$

Where $K_{1c}$ is the material fracture toughness. The Forman correction is more generalised and could be used, in some cases, when tests under different R values also show FCG curves slope with various slopes, m.

**Stress intensity Threshold values ($\Delta K_{th}$)**

One of the major deficiencies of the FCG equation is its dependency on an accurate definition of $\Delta K_{th}$. It is assumed that below this value cracks would NOT propagate. The threshold values are difficult to obtain experimentally and usually data is extrapolated. The R-ratios have a strong effect on the threshold values. Other factors influencing the threshold values are the crack closure and the loading rate.

For example, the fatigue $\Delta K_{th}$ of low strength steel can be as low as 3 MPa $m^{1/2}$ For high R-ratio of 0.75 while its $K_{1c}$ value could be as high as 300 MPa $m^{1/2}$. This is in particular important in situations of high frequency and low load, for example rotation of a high-speed rotor. In practice, usually a cut-off stress level (or gate) is applied in the analysis and empirical-experimental approach must be applied.

**Crack closure**

Our discussion so far was referred mainly to the assumption that each cycle open the crack-tip. However, it is generally accepted that crack closure **decreases the FCG rate.**

Elber introduced one of the simplest models that account for crack closure measuring crack closure in 2024 -T3 aluminium alloy. He made the assumption that only the portion of $\Delta K$ in which the crack is open contributes to the FCG. An expression for the stress intensity in which the crack just opens was introduced, $K_{op}$.

Below is a simple empirical relation to obtain an effective stress intensity range that takes into account the closure as a function of the stress ratio R based on the Elber model.

$$\Delta K_{eff} = K_{max} - K_{op} = K_{max} - \Delta K\left(\frac{1}{1-R} - 0.5 - 0.4R\right) \qquad (-0.1 \le R \le 0.7)$$

Where ΔK is the applied stress intensity range.

However, the closure also effects the threshold values, $\Delta K_{th}$, and high closure INCREASE the $\Delta K_{th}$. A simple modification, using Elber work gives:

$$\Delta K_{th/eff} = \Delta K_{th}(0.5 + 0.4R)$$

FCG constants for several materials are given below:

| Material | $\Delta K_{TH}$ ($MN\,m^{-3/2}$) | $m$ | $C^*$ ($\times 10^{-11}$) |
|---|---|---|---|
| Mild steel | 3.2–6.6 | 3.3 | 0.24 |
| Structural steel | 2.0–5.0 | 3.85–4.2 | 0.07–0.11 |
| Structural steel in sea water | 1.0–1.5 | 3.3 | 1.6 |
| Aluminium | 1.0–2.0 | 2.9 | 4.56 |
| Aluminium alloy | 1.0–2.0 | 2.6–3.9 | 3–19 |
| Copper | 1.8–2.8 | 3.9 | 0.34 |
| Titanium | 2.0–3.0 | 4.4 | 68.8 |

* Units of $C$ will give $da/dN$ in m/cycle when $\Delta K$ is in $MN\,m^{-3/2}$.

**Life estimates using FCG - constant amplitude load**

The FCG equation is based on the rate of crack propagation per cycle and the stress intensity range, ΔK. Since ΔK increases with increase in crack length we have to integrate between the number of cycles to failure, $N_f$, and the initial number of cycles, $N_i$, or similarly between the respective crack lengths, $a_f$ and $a_i$.

$$\int_{N_i}^{N_f} dN = N_f - N_i = N_{if} = \int_{a_i}^{a_f} \frac{da}{f(\Delta K, R)}$$

To perform the integration for a particular case a specific equation for R and ΔK is required, while some useful closed form solutions exists it is necessary in many cases to perform a numerical integration.

For a particular R-ratio and assuming that the geometrical factor β is independent of the crack length (incorrect in many cases) we can integrate the equation above:

$$N_{if} = \int_{a_i}^{a_f} \frac{da}{C(\Delta K)^m} = \int_{a_i}^{a_f} \frac{da}{C\left(\beta \Delta S \sqrt{\pi a}\right)^m} da = \int_{a_i}^{a_f} \frac{da}{C\left(\beta \Delta S \sqrt{\pi}\right)^m} \frac{1}{a^{(m/2)}} da$$

If we assume that C, β, ΔS and m are all constants, the only variable is ‘a’ the crack length and integration is straightforward, giving:

$$N_{if} = \frac{a_f^{(1-m/2)} - a_i^{(1-m/2)}}{C\left(\beta \Delta S \sqrt{\pi}\right)^m (1 - m/2)} \qquad (m \neq 2)$$

It is possible to show that this closed form integration is highly sensitive to the value of the assumed initial defect, $a_i$ but less sensitive to the final one, $a_f$. This is a clear limitation of any FCG analysis: IT IS HIGHLY SENSITIVE TO THE ASSUMED INITIAL CRACK LENGTH, $a_i$.

**Note about the units:** To avoid difficulty with units in calculating FCG life the units quantities should correspond to the correct $\Delta K$ units. For example if [MPa $m^{1/2}$] is used for $\Delta K$ , C should correspond to [m/cycles], stress to [MPa] and crack length, to [m].

**Example (FCG and R correction)**

A centre-cracked steel plate has dimensions of W=76mm and t=6mm and it contains initial crack of length 2a=2mm. It is subjected to tension constant cyclic loading between load values of $P_{min}$=80kN and $P_{max}$= 240kN.
(Material properties: $\sigma_y$=1255 MPa, $K_{1c}$=130 MPa $m^{1/2}$, $C_{(R=0)}$= 5.11x$10^{-10}$ mm/cycle, m=3.24, $\gamma_{(Walker)}$=0.42)

1. At what crack length $a_f$ failure is expected and is it due to plastic yield or fracture?

For fully plastic yield conditions the crack length can be estimated for this geometry as (b = W/2):

$$\boxed{a_{fy} = b\left(1 - \frac{P_{max}}{2bt\,\sigma_y}\right)}$$

Inserting values we obtain:

$$a_{fy} = 38\left(1 - \frac{240000}{2x38x6x1255}\right) = 22.1mm$$

To obtain the fracture critical crack length we use the fracture toughness equation. The maximum far field stress in the plate is:

$$S_{max} = \frac{P_{max}}{2bt} = \frac{240000}{2x38x6} = 526 \text{ MPa}$$

Assuming that the geometry factor, $\beta$, is constant and is equal to 1:

$$a_{fc} = \frac{1}{\pi}\left(\frac{K_{1C}}{\beta S_{max}}\right)^2 = \frac{1}{\pi}\left(\frac{130}{1x526}\right)^2 = 0.0194m = 19.4mm$$

We have a fracture failure since 19.4 < 22.1. However, the real value of $\beta$ is crack length dependent. This would make the estimated final crack at fracture shorter.

Make some iterations to obtain a more accurate fracture crack length.

2. Estimate the number of cycles to fracture.

We use the Walker model for the calculations at a specific R-ratio:

$$R = \frac{S_{min}}{S_{max}} = \frac{P_{min}}{P_{max}} = \frac{80}{240} = 0.333$$

At this point it is convenient to calculate an R dependent intercept C value since it is assumed that the exponent m is constant, after manipulation of the Walker equation, we obtain:

$$\boxed{C = \frac{C_1}{(1-R)^{m(1-\gamma)}}}$$

Where $C_1$ is the intercept for R=0

$$C = \frac{5.11x10^{-10}}{(1-0.333)^{3.24(1-0.42)}} = 1.094x10^{-9}\,[mm/cycle] = 1.094x10^{-12}\,[m/cycle]$$

The stress range is:

$$\Delta S = S_{max}(1-R) = 526(1-0.333) = 351 \text{ MPa}$$

Finally, to estimate the number of cycles to failure we assume an average $\beta$ value of 1.05 during the FCG and use the closed form solution for the integration from $a_i$ to $a_f$:

$$N_f = \frac{a_f^{(1-m/2)} - a_i^{(1-m/2)}}{C\left(\beta\Delta S\sqrt{\pi}\right)^m (1-m/2)} = \frac{(0.0194)^{-0.62} - (0.001)^{-0.62}}{1.094x10^{-12}\left(1.05x351x\sqrt{\pi}\right)^{3.24}(-0.62)}$$

$$N_f = 68x10^3 \text{ cycles}$$

Where (1-m/2) = -0.62

Note that this is an initial estimate and a refinement should be carried out numerically with geometry factor $\beta$ dependency on the crack length a.

**Example (in class exercise)**

A service engineer has detected a 2mm long edge crack in aerospace component made of steel alloy that has a fracture toughness $K_{1c}$ = 82.5 MPa $m^{1/2}$ and yield stress $\sigma_y$ = 750 MPa. The component is subjected to a cyclic tension (R= 0) stress of $\Delta\sigma$ = 140 MPa and the material Paris-type FCG relation is:

$$\frac{da}{dN}(m/cycles) = 0.66x10^{-8}\left(\Delta K\left[MPa\sqrt{m}\right]\right)^{2.25}$$

Estimate how many more cycles can this component endure before failure if for this crack geometry $\beta$ = 1.12.

**Solution:**

1. Calculate the final crack length.
2. Calculate the number of cycles using direct integration formula.

(Answer: 3150 cycles)

**Life estimates using FCG - variable amplitude load**

In cases of variable load the crack growth rate, da/dN may vary from cycle to cycle, depending upon $\Delta K$ and R of the cycle involved and also the sequence of the cycles. In general two approaches often used:

**1. Cycle by cycle analysis**
Summation of crack increments for every cycle by taking into account factors such as history effects for example overload and retardation. This could take much computer time since the growth rate must be calculated cycle by cycle.

**2. Block cycles analysis**

Accumulation of the cycles using, for example, rainflow counting and ignoring the history effects and calculating crack growth by adding the growth of each block of cycles (defined by stress range and stress ratio). This approach is fast and usually applied initially to estimate the cycles damage.

**Cycle by cycle procedure:**

This is in fact a numerical integration of da/dN but since dN = 1 we calculate the crack increment, Δa, every cycle and add it to the total crack:

$$a_{j+1} = a_j + \Delta a_j = a_j + \left(\frac{da}{dN}\right)_j \qquad \text{for example} \quad \left(\frac{da}{dN}\right)_j = C\left(\frac{\Delta K}{(1-R)^{1-\gamma}}\right)_j^m$$

Assuming Walker correction. Note that C, m and $\gamma$ are material constants while $\Delta K$ and R are cycle dependent.

The crack length after the N cycles will be:

$$a_N = a_j + \Delta a_j = \sum_{j=1}^{N} \Delta a_j = \sum_{j=1}^{N} \left(\frac{da}{dN}\right)_j$$

**Block cycles analysis:**

The crack growth for each block of cycles is:

$$\Delta a_{block} = N_{block} \left(\frac{da}{dN}\right)_{block}$$

And the total crack lengths:

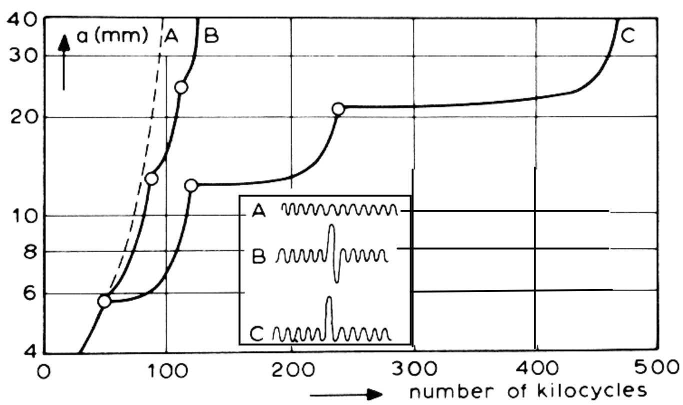
$$a_N = \sum_{1}^{Total} \Delta a_{block} = \sum_{1}^{Total} N_{block} \left(\frac{da}{dN}\right)_{block}$$

In practice, the total load history is usually unknown and therefore a mixed approach is used. Applying repetitions of predicted load history in a specific sequence.

**Sequence effects (retardation)**

Unlike the random loading life to crack initiation summation procedure, where the order of the cycles was assumed unimportant (Miner's rule) the sequence of the applied load has a clear effect on the rate of crack growth.

For example, a high tension overload decreases the growth rate, this is because the overload increase the crack-tip plastic zone and more cycles needed before the crack grows beyond this region. This beneficial effect is called *crack growth retardation* and is illustrated in a centre cracked plate made of 2024 T3 aluminium below:




A – No overload

B – Overload, R= -1

C – Overload, R=0

Modelling of the retardation process is complex and is not well defined (yet) however some simple models exists for example, **Wheeler** introduced a retardation factor, $\phi_r$, to obtain a modified FCG rate due to overload as follows:

$$\left(\frac{da}{dN}\right)_{Retard} = \phi_R \frac{da}{dN}$$

$$\phi_R = \left(\frac{\Delta a + r_{p,current}}{r_{p,overload}}\right)^{\delta}$$

Where $r_{p,current}$ is plastic zone for the current cycle and $r_{p,overload}$ is the plastic zone for the overload cycle. $\delta$ is, in general, a material property.

**Units conversion**

It is customary in the literature to use the empirical units for fracture measurements and calculations.

Conversion of the stress intensity K from Ksi in$^{1/2}$ to MPa m$^{1/2}$ or vice versa is:

$$1 \text{ Ksi in}^{1/2} = 6.89\text{MPa } (0.0254\text{m})^{0.5} \approx 1.09 \text{ MPa m}^{1/2}$$

and:

$$1 \text{ MPa m}^{1/2} \approx 0.92 \text{ Ksi in}^{1/2}$$

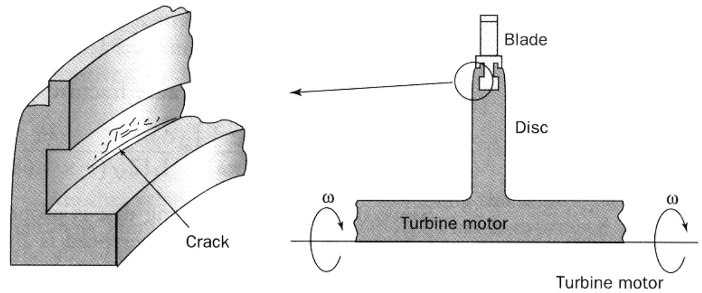
**Example (in class exercise)**

A turbine blade is fitted into aluminium alloy disc on a rotor circumference disc as shown in the Figure overleaf.
During assembly a 0.1mm scratch was introduced into the surface of the disk. The rotation of the turbine causes a tension cyclic stress of 350 MPa at the plane of the scratch.

- Estimate the number of stress cycles to fracture failure of the disk.
- Calculate the maximum allowed initial crack if the rotor is to undergo 2000 cycles during its operation.
- By what factor the rotation speed has to be increased (over-speed) for catastrophic failure to occur in service?

(Note: the stress is proportional to the square of the rotation speed).

(Material properties: $K_{1c}$=35 MPa m$^{1/2}$, $C_{(R=0)}$= 4x10$^{-11}$ m/cycle, m=3.54, assume: R=0, $\beta$=1.12)




Solution procedure:

1. Estimate the final crack length using fracture toughness. Answer: 2.54mm
2. Estimate the maximum number of the cycles to failure. Answer: 3117 cycles.
3. Calculate the initial crack length using the service cycles. Answer: 0.168mm.
4. Use this critical crack to obtain the fracture stress.
5. Calculate the root square of the stress ratio, maximum to in-service stress. Answer: 2.

**Example**

A mild steel plate is subjected to two different cyclic fatigue load tests as follows:

1. A constant amplitude test, $\sigma_{max} = 250$, $\sigma_{min} = -25$ MPa.
2. Same but with one overload cycle of twice the maximum stress after 50,000 cycles.

Estimate the number of cycles to failure of the plate using FCG considerations and assuming initial edge crack of 0.5mm in an infinite wide plate.

(Material properties: $K_{1c}$=100 MPa $m^{1/2}$, $C_{(R=0)}$= 6.9x$10^{-12}$ m/cycle, m=3, geometry factor $\beta$=1.12, $\sigma_y$=500 MPa)

**Solution:**

1. We assume that we can neglect the compressive stresses. Therefore R = 0 and the stress range is 250 MPa.

The final crack length is therefore:

$$a_{fc} = \frac{1}{\pi}\left(\frac{K_{1C}}{\beta S_{max}}\right)^2 = \frac{1}{\pi}\left(\frac{100}{1.12x250}\right)^2 = 0.0406m = 40.6mm$$

And the total number of cycles to failure assuming β is constant:

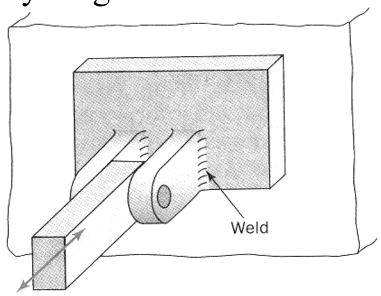
$$N_f = \frac{a_f^{(1-m/2)} - a_i^{(1-m/2)}}{C\left(\beta\Delta S\sqrt{\pi}\right)^m (1-m/2)} = \frac{(0.0406)^{-0.5} - (0.0005)^{-0.5}}{6.9x10^{-12}\left(1.12x250x\sqrt{\pi}\right)^3(-0.5)} = \frac{-39.7584}{-4.2171x10^{-4}}$$

$$N_f = 94278 \text{ cycles}$$

2. We use the Wheeler model for crack retardation and hence need to calculate the crack tip plastic zone before and after the large cycle was applied, taking δ = 1.8 plane stress. (Solution to be given in class).

**Tutorials**

1. A support bracket is welded to a plate as shown. A cyclic tension load at the lug causes stress cycles of +/-50 MPa at the weld. Calculate the maximum size of defect if the stress intensity range threshold value is, $\Delta K_{th}$=1.65 MPa $m^{1/2}$.




(Answer: 0.27mm).

2. If da/dN = $1x10^{-6}$ m/cycles at $\Delta K$ = 10 MPa $m^{1/2}$ and da/dN = $1x10^{-3}$ m/cycles at $\Delta K$ = 100 MPa $m^{1/2}$ (R=0).

a. Determine Paris equation constants (Answer: $10^{-9}$, 3).

b. Convert to inch/cycles and ksi $in^{1/2}$ and determine the Paris equation constants.

3. Experimental programme was carried out on a batch of aviation grade aluminium 2024-T3 Al at two different Ratios and the following Paris equations were derived:

$$\frac{da}{dN} = 1.60x\,10^{-8}(\Delta K)^{3.59} \qquad R = 0.1$$

$$\frac{da}{dN} = 3.15x\,10^{-8}(\Delta K)^{3.59} \qquad R = 0.5$$

Determine the value of the intercept C and the exponent $\gamma$ that can describe this behaviour by the Walker equation. (Answer: $1.42x10^{-8}$, 0.679 and Hint: only C is a function of R)

4. Using crack length of a = 0.1 inch, $\sigma_{max}$ = 22 ksi and R=0.2 and using Paris equation constants obtained in 2 above calculate: ($K_{1c}$=100 MPa $m^{1/2}$, $\gamma$ =0.6)

a. Crack growth rate predicted by the Walker equation. Answer: 0.0023mm/cycles)

b. Crack growth predicted by the Forman equation. (comment on this result) (Answer:$2.127x10^{-5}$ mm/cycles)